

What Determines a Simulation Benchmark Ranking? Structural Blindness, Finite Configuration Grids, and Evaluation Budget in Simulation-Based Policy Comparison

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Abstract

Simulation benchmarks increasingly rank robot foundation policies, treating a calibrated simulator as evidence of which is better. We ask which ranking claims a benchmark’s own evidence determines, on a widely used sim-to-real suite across two embodiments, enumerating every grid. First, the calibration protocol is structurally blind: replaying 98 demonstrations identifies the controller ratio and delay, while a common gain rescaling moves the replay objective by at most 3×10^{-6} of its own composite units over a sixteenfold range — against 1.4×10^{-3} between two independent implementations of the same dynamics — and halving the torque limit leaves 97 of 98 replays bitwise unchanged. The same test rejects the simulator’s shipped setting; a full census at the setting it prefers moves the point-calibration verdict from abstention to a declaration on a difference change of 0.031 that is not itself resolved, while the set-valued verdict abstains at both points. Second, the benchmark is a finite configuration population: the episode index fixes the initial state modulo 24 to 300 configurations, and two ports enumerate different grids. Enumerating one zeroes configuration sampling error; among the terms that remain is the suite’s averaging over four robot texture variants, which carry no physics yet span 0.400 in a rate, making the published number a marginal rather than the rate at any one. Third, following the published protocol reproduces its rates to 0.038 mean absolute difference, yet its own 72-episode budget resolves only one or two of four orderings. A set-valued verdict bounds only the ambiguity the compatible parameters express: on the pair we selected from the 5 of 62 published pairs whose real and simulated means point opposite ways, it declares the ordering those means reverse. Four of our claims are retracted.

Keywords: Policy evaluation, Sim-to-real transfer, Identifiability, Partial identification, Benchmark reproducibility, Robot manipulation

1 Introduction

A simulator calibrated against real demonstrations is increasingly treated as evidence about which robot policy is better. Simulation suites that mirror real evaluation scenes (Li et al., 2025) are used to compare foundation policies at a scale no physical laboratory can match, and their published tables are read as rankings: this policy beats that one on this task. The appeal is obvious. A physical evaluation of a single policy on a single task costs hours of robot time and is difficult to reproduce; a simulated one costs minutes and is, in principle, exactly repeatable.

The standard question asked of such a suite is how well it correlates with reality, and it is usually answered with a correlation coefficient or a rank agreement over a handful of policies. This paper asks a question that comes before that one, and that can be answered without a robot:

Given the evidence actually used to build and run a simulation benchmark — the demonstrations it was calibrated on, the configurations it enumerates, and the episodes it runs — which ranking claims does that evidence determine?

The distinction matters because a correlation study treats the simulator’s output as a measurement with some error attached, whereas our question asks whether the output is a well-defined function of the inputs at all. If two simulator configurations are equally consistent with every piece of calibration data and they disagree about which policy wins, then no amount of correlation analysis repairs the ranking: the evidence does not contain the answer. This is the distinction between *identifiability* and *statistical power*, and one of our findings is that practitioners systematically conflate them — including, at two points in this project’s history, us.

We answer the question on a widely used sim-to-real evaluation suite for manipulation, across two embodiments (a WidowX bridge setup and a Google-robot fractal setup), eight policy configurations across three model families — Octo (Ghosh et al., 2024), OpenVLA (Kim et al., 2025) and RT-1 (Brohan et al., 2023) — and a complete enumeration of every task’s configuration grid. Three findings, each independent of the others.

The calibration protocol is structurally blind (Sect. 4).

The protocol identifies some dynamics parameters and is, over the range it excites, uninformative about others. Replaying 98 real demonstrations, we show analytically and then empirically that a common scaling of the joint PD gains and the value of the torque limit leave the replay trajectory unchanged to within magnitudes we quantify below — the first because the free-space response depends on the ratio d/k up to inertial terms that these slow, teleoperated demonstrations do not excite, the second because commanded torque never saturates along demonstrated velocities. Sect. 4.2 states both as insensitivity over a measured domain rather than as exact invariance, and quantifies the residual. The measured effect is 0.141 mm on one simulator stack and 0.207 mm on a second, independent one for the gain scaling. For the torque limit the replayed trajectory is bit-wise identical on 97 of the 98 episodes, on both stacks; the single exception is the same demonstration on each, differing by 0.149 μm on one stack and 0.050 mm on the other. That pattern is what the saturation argument predicts — invariance holds exactly where the commanded torque stays inside the limit, and one demonstration evidently approaches it — so we state the invariance as near-exact rather than exact. These are structural invariances of the protocol, not flat regions of a noisy objective, so no improvement in fitting technique removes them.

The benchmark is a finite configuration population, not a sample (Sect. 5).

The episode index determines the initial state modulo the size of the task’s configuration grid — 24 configurations for three bridge tasks, 64 for one port’s eggplant task, 300 for the fractal coke-can task. Episodes beyond that count are exact repeats, so the common practice of running more episodes stops buying information about configurations at a threshold most users never learn. Two widely used ports of the same suite enumerate *different* grids, so a number computed on one is not comparable to a number computed on the other. Enumerating the grid drives configuration sampling error to exactly zero, and what remains is instructive: policy-seed noise, implementation-build drift, and the suite’s own averaging over four

robot *texture* variants, across which one policy’s success rate ranges over 0.400 — that is, 40 percentage points, the unit in which every effect size in this paper is quoted. These three are not comensurable and we do not rank them: they are a standard deviation, a paired shift and a range over four settings. The texture parameter changes the colour of the robot model, carries no physics whatsoever, and is invisible to replay calibration for exactly that reason; the averaging over it is part of the documented protocol, and what we contribute is its magnitude and the observation that it makes the published number a rate marginal over the four variants, which is a different estimand from the conditional rate a single-variant reproduction measures.

The published protocol does not resolve the rankings it reports (Sect. 6).

We follow the reference protocol, including its pseudo-random-number lifecycle, and reproduce the published success rates to a mean absolute difference of 0.038 across eight cells, and on the second embodiment to +0.003 and −0.040. The differences are one-signed — six of the eight cells fall below the published value, two match it, none exceeds it, mean −0.038 — and we report that because our own first reproduction did not show it: that run re-seeded the policy every episode, which inflated its variance and let a noisier estimator look like a closer match (Sect. 6.2). We do not convert those eight cells into a platform constant; they share tasks, policies and seeds. What licenses the rest of the paper is narrower and sufficient: every effect we report is a difference between two policies measured in the same configuration on one pipeline and one build, where an offset common to both policies cancels. That last condition is not decoration: Sect. 5.5 measures a change of build shifting $\hat{\Delta}$ itself by −0.109, so a build change is *not* a common offset, and we never compare across one. At the protocol’s own 72-episode budget, one of the four policy orderings it reports is statistically resolvable under the variance model the protocol’s own randomisation implies, and two under a coarser model that treats each run rather than each configuration as the replication unit; we report both and argue for the former (Sect. 6.3). On carrot the published difference is +0.014 and ours is +0.028 with interval

$[-0.039, +0.094]$ under either: a number whose sign its own budget does not determine, which is precisely the failure mode this paper is about.

Having established that ranking claims are underdetermined from three independent directions, we ask what can be done about it (Sects. 7 and 8). The natural remedy is a set-valued verdict: declare an ordering only when it holds for every simulator parameter the calibration data cannot exclude. We measure that criterion’s reach, and report both halves of the answer. It is conservative as designed on synthetic ground truth, where the point criterion’s error rate *rises* with episode count to 0.88, and it stabilises verdicts on real replicate data. At the reference protocol’s three seed sets it disagrees with point calibration on 3 of 17 policy pairs, and at a single seed set on 2 of the 13 pairs one seed set can evaluate at all — but when we spend more seeds, two of the three disagreements turn out to have been our own insufficient power, leaving one. And on the single policy pair for which published real-robot results contradict the simulator, the criterion over the invisible fibre *declares the ordering the published real means reverse*. Its reach stops at the parameter family the calibration data can constrain, and the sim-to-real gap on that pair lies outside it.

1.1 Contributions

This is an audit of an evaluation pipeline, and its content reduces to three claims, each of which can be checked or refuted on its own from the released records.

C1. The calibration protocol is weakly informative about directions that move policy gaps and declarations. *Estimand:* the benchmark’s own pairwise success-rate difference $\hat{\Delta}$. *Scope:* two simulator stacks, the ranges the policy sweeps use, 98 teleoperated free-space demonstrations. Demonstration replay pins the gain ratio and the execution delay tightly and says almost nothing about a common gain rescaling or an inactive torque limit; those insensitive directions shift $\hat{\Delta}$ by up to 0.121 and move declarations to abstentions. We claim the shifts and the verdict changes, *not* a reversal of any sign-determined ranking — we found none, and say so in the limitations. *Refutable by* showing the loss is sensitive to those directions over the ranges we tested, or that the shifts are smaller than reported.

C2. The evaluation target is a finite population, so replicating one setting cannot remove the difference between settings.

Estimand: the same $\hat{\Delta}$, at a stated operating point and a stated variant. *Scope:* the three bridge tasks and pick-coke-can. The configuration set is index-determined and enumerable rather than sampled, so enumerating it sets configuration sampling error to exactly zero. What then remains divides in two, and the claim concerns only the second part: policy-seed noise *does* fall with more replicates of the same setting, and we spend budget on it; but the implementation build, the calibration-invisible parameters and the physics-free texture variant are differences *between settings*, and no number of replicates at one setting bounds them. We do not claim that all residual uncertainty is budget-insensitive. *Refutable by* showing the grid is not index-determined, or that a between-setting difference we report shrinks when the same setting is replicated.

C3. At the budgets in use, individual pairwise claims are decided by modelling choices the evidence does not settle.

Estimand: the verdict — declare or abstain — on one pairwise claim. *Scope:* 17 bridge pairs, one to five seed sets. Three distinct things belong here and we keep them apart. (i) *Modelling choices the data cannot adjudicate:* the protocol’s own 72-episode budget resolves one or two of the four orderings it reports depending on the variance model, and the observation unit — runs or directory means, both consistent with the stated formula — moves two verdicts among the 18 rows of the release’s core table (the 17 bridge pairs plus the pick-coke-can pair). (ii) *A choice the evidence does settle, against the benchmark:* the controller operating point, which the replay objective rejects, moves the point-calibration verdict from abstention to a declaration; here the evidence has a preference and the benchmark is not at it. The set-valued verdict abstains at both points. (iii) *Neither of the above:* the pseudo-random-number lifecycle, where the reference code fixes the correct protocol and our first re-seeding implementation was simply an error we have withdrawn; we keep it in the paper as a reproducibility lesson, not as evidence for (i). *Refutable by* showing that a choice under (i) is determined by the evidence, or that the verdicts are stable across it.

A set-valued verdict over the insensitive directions is the natural response to C1, and we report what it buys and where it stops rather than as a contribution in its own right: it is classical partial identification applied as an audit tool, it carries no coverage guarantee as we compute it (Sect. 3.2), and it does not protect against the operating-point choice of C3. Four conclusions of our own are withdrawn along the way and are listed in Sect. 1.2.

Where each claim is established, and the measurements that carry it:

C1 Sect. 4 argues the non-identifiability from the mechanism and measures it on two independent simulator stacks; Sect. 7 gives the decision-relevant instance, a calibration-invisible torque limit that displaces a policy difference by 0.121 and moves verdicts from decidable to abstaining; Sect. 7.7 re-runs the complete census at the operating point the replay objective prefers, on the same configurations and the same policy seeds.

C2 Sect. 5 identifies the benchmark as a finite configuration population, gives the configuration count per task, shows that two widely used ports evaluate different benchmarks, and sets out an estimand taxonomy with an uncertainty budget for the terms a census leaves. The sharpest of those terms is the suite’s documented averaging over four robot texture variants — a parameter carrying no physics, across which one policy’s rate spans 0.400 — which makes the published number a rate marginal over the four, so a single-variant reproduction targets the conditional rate and can differ by -0.173 to $+0.227$ with nothing wrong in either.

C3 Sect. 6 reproduces the published protocol down to its pseudo-random-number lifecycle, and that lifecycle turns out to be consequential: re-seeding the policy each episode instead of advancing one stream doubles the across-run spread of a cell’s rate (0.052 to 0.109), moves the mean difference from the published rates from -0.005 to -0.038 , and decided the sign of one comparison. Sect. 8 measures the verdict ladder’s coverage and power on synthetic ground truth and its stability on real replicates, and reports the negative result that delimits it: on a pair selected from the 5 of 62 published pairs whose real and simulated means contradict each other, the criterion declares the ordering the real means reverse.

The appendices carry the reproducibility assets: the original reference stack running headless on a GPU-less host through a purpose-built Vulkan compatibility layer, the configuration-census protocol, and append-only provenance logs for every episode reported here.

1.2 What we withdraw

Four conclusions of our own are retracted in this paper, each after we ran the experiment designed to test it.

1. *“Eggplant nominal: octo-small beats octo-base.”* Declared from one seed set with a per-episode bootstrap, interval $[+0.08, +0.32]$. Three same-generation seed sets give $+0.141$, -0.031 and -0.047 . Withdrawn (Sect. 5.5).
2. *“The torque effect replicates across two seed sets.”* The episode index fully determines the configuration, and the policy in question decodes deterministically, so the two “seed sets” were repeat runs of identical rollouts. Withdrawn and replaced by a configuration-census protocol (Sect. 5.3).
3. *“Budget cannot explain the point-versus-set disagreements.”* Taking the census from three to five seed sets resolved two of the three disagreements, and dissolved the one between-condition sign reversal we had claimed. Withdrawn; Sect. 7 states the weaker conclusion the data supports.
4. *“A faithful re-run of the published protocol reverses a published ordering.”* Our reproduction placed octo-base above octo-small on carrot where the published table has the reverse. It was not faithful: the reference seeds its policy once per run and lets one stream advance, while we re-seeded at every reset, so all 24 configurations in a run shared one noise realisation. Re-run under the reference lifecycle, carrot agrees in sign with the published table and no cell disagrees. Withdrawn (Sect. 6.3); the budget claim that motivated it survives in the weaker form that carrot’s published $+0.014$ has an interval of $[-0.039, +0.094]$.

Four further ranking-flip candidates seen at 24 and 48 episodes were withdrawn before being claimed, when adding episodes removed them. We list this in the body rather than in a limitations section because a reader’s estimate of how much

to trust our surviving claims should be calibrated by how the non-surviving ones died.

1.3 Scope

We make no claim about which policy is actually better on any real robot, and we ran no hardware ourselves. Where real-robot ground truth is needed — in Sect. 8.4’s delimiting experiment — we use the success rates published alongside the benchmark for the same policies and task, which are the reference values the benchmark itself is validated against. Two public evaluation services that would have provided independent hardware results were unavailable during this work: one endpoint was offline throughout and the other’s queue allocation could not be confirmed. Our claims are about what the simulated evidence determines, which is a question about the benchmark rather than about the robot.

2 Related work

Simulation-based evaluation of manipulation policies.

Recent evaluation suites construct simulated scenes that mirror real evaluation setups, calibrate the simulator against real demonstrations, and report success rates intended to be comparable with real-robot numbers (Li et al., 2025). Parallel efforts pursue automated real-robot evaluation (Zhou, Atreya, Tan, Pertsch, & Levine, 2025) and distributed evaluation across laboratories (Atreya et al., 2025). All are motivated by the cost and irreproducibility of physical evaluation. What validates each of them differs and the difference matters here: SIMPLER reports rank correlation and a mean maximum rank violation against published real rates, AutoEval automates the real evaluation itself and so validates against its own hardware runs, and RoboArena aggregates distributed real evaluations rather than validating a simulator at all. Our work takes such a suite as its object of study rather than as a tool, and asks not how closely it agrees with reality but which of its own outputs are determined by its own inputs.

We should be precise about the suite we study, because it anticipates more than a correlation analysis. Li et al. (2025) discuss the weaknesses of Pearson correlation for this purpose,

note that small real differences make rank agreement unstable, and propose a mean maximum rank violation to measure ordering agreement directly. What we add is not the observation that correlation is a poor summary. It is a per-pair analysis conditioned on a stated calibration protocol and a stated evaluation budget: which individual ranking claims that evidence determines, where the parameter directions the protocol cannot see enter, and how the answer moves with the operating point and the observation unit.

Confidence statements for simulated evaluation, and what we do not offer.

The closest neighbours are methods that attach a finite-sample statement about *real* performance to a simulated or partly-simulated evaluation. [Badithela et al. \(2025\)](#) use a small set of paired real and simulated evaluations with prediction-powered inference to give intervals for real performance, and [Zhu et al. \(2026\)](#) give scenario-conditioned predictions with conformal calibration. [Jain et al. \(2025\)](#) belongs beside these but is not the same kind of thing: it is a real-to-sim reconstruction and evaluation pipeline validated by empirical correlation with real evaluations, not a method that carries a finite-sample interval, and we separate it here so the grouping does not imply a guarantee it does not claim. The distinction from our work is about the estimand and the information required, and it is not in our favour on the question the first kind of method answers. They target real performance and pay for it with paired real data. We have no paired real evaluations of our own and therefore make no claim about real performance: our estimand is the benchmark’s own quantity, and our output is which of *its* pairwise claims its own evidence supports. A reader wanting an interval on real performance should use a method of that first kind; what this paper offers is an audit of what the simulated evidence settles before any sim-to-real correction is applied. We cannot present a fair measured comparison against them, because doing so needs the paired real data they assume and we lack, and we prefer to state that than to construct a baseline they would not recognise.

Identifiability and system identification.

That not every parameter of a dynamical model can be recovered from a given excitation is classical, and structural identifiability analysis is standard in system identification ([Bellman & Åström, 1970](#); [Ljung, 1999](#)). Robot calibration practice generally acknowledges it in the form of excitation design: trajectories are chosen to make the parameters of interest observable ([Swevers, Ganseman, Tükel, De Schutter, & Van Brussel, 1997](#)). The gap we address is that demonstration-replay calibration does not get to choose its excitation — the demonstrations are whatever humans happened to teleoperate, they are free-space and slow, and the resulting blindness is inherited rather than designed. We treat the blind directions not as a nuisance to be reduced but as a set to be propagated, which connects to the next literature.

Partial identification and set-valued inference.

When data do not pin down a parameter, one can report the set of values consistent with the data and the corresponding set of implied conclusions, rather than a point estimate and a confidence interval around it ([Manski, 2003](#); [Tamer, 2010](#)). Model-confidence-set procedures provide the specific device we use: invert a test to obtain the collection of models the data cannot distinguish, then require a conclusion to hold across that collection ([Hansen, Lunde, & Nason, 2011](#)). Our contribution is not methodological novelty here but instantiation and measurement — we construct the compatible set for a real calibration protocol, propagate it to policy rankings, and quantify what the resulting criterion does and does not buy, including a case where it fails.

Randomising what the simulator gets wrong.

A large literature responds to simulator inaccuracy by randomising over it during training, so that the learned policy is robust to whichever value is right — over visual appearance ([Tobin et al., 2017](#)) and over dynamics ([Peng, Andrychowicz, Zaremba, & Abbeel, 2018](#)). Our subject is the mirror image of that programme. Randomisation is a technique for making a *policy* insensitive to parameters the modeller cannot pin down; we ask what remains of an *evaluation* when those same

parameters cannot be pinned down. The two interact in a way worth stating plainly, because one of the terms that survives a configuration census on our data is a purely visual one: it is not news that visual variation matters for transfer, which is precisely the premise of domain randomisation. Nor is the aggregation over it undocumented — the suite reports it as part of its protocol (Li et al., 2025). What we add is the magnitude, and the observation that averaging over such a variant makes the published number a rate *marginal over* the variants rather than the rate at any one of them, so a single-variant reproduction targets a different quantity (Sect. 5.5). That is a reporting and estimand point, not an argument against randomisation.

Reproducibility and statistical practice in learned-policy evaluation.

Sensitivity of reinforcement learning results to random seeds, and the resulting unreliability of small-sample comparisons, is documented and has produced concrete recommendations on reporting (Agarwal, Schwarzer, Castro, Courville, & Bellemare, 2021; Colas, Sigaud, & Oudeyer, 2018; Henderson et al., 2018). Our Sect. 5 contributes two items this literature does not cover for simulated robot benchmarks: the initial-state grid is finite and enumerable, so the configuration axis is a population rather than a sample and its sampling error can be driven to exactly zero; and the *implementation build* — the port branch plus the inference stack version — behaves as a parameter of the same order as the ones being studied. The first changes which bootstrap is correct; the second means that even a perfectly specified bootstrap understates the uncertainty of a number copied from a published table.

3 Problem setup

3.1 Notation

Let z denote the simulator’s physical parameters (joint PD gains, torque limits, contact friction and density), and let π denote the *implementation build*: the port branch, the renderer path and the policy inference stack version. A task provides a finite set $\mathcal{C}$ of initial configurations. A policy i run from configuration c under parameters z and

build π with internal random source σ either succeeds or fails, written $s_i(c, z, \sigma; \pi) \in \{0, 1\}$. The simulated difference between two policies is

$$\Delta_{ij}^{\text{sim}}(z; \pi) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} [\mathbb{E}_{\sigma} s_i(c, z, \sigma; \pi) - \mathbb{E}_{\sigma} s_j(c, z, \sigma; \pi)], \quad (1)$$

and $\Delta_{ij}^{\text{real}}$ denotes the corresponding real-robot difference. Three properties of this definition carry the weight of Sect. 5 and are stated here because they are easy to overlook: the sum over $\mathcal{C}$ is over a *population*, not a sample; σ is degenerate for deterministically decoding policies, so for those the benchmark value is measurable exactly; and $\mathcal{C}$ itself depends on π , so two builds with different grids yield two different estimands that cannot be paired per episode.

3.2 The compatible rule, and why we do not call it a confidence set

Calibration observes real demonstrations and scores a candidate z by a replay loss $L(z)$. Rather than reporting the minimiser $\hat{z}$, we invert a one-sided test at level α : a parameter setting is *compatible* if its per-demonstration error is not significantly worse than the best,

$$C_{\alpha} = \left\{ z : \text{lo}_{1-\alpha}(\mathbb{E}[L(z) - L(\hat{z})]) \leq 0 \right\}, \quad (2)$$

where $\text{lo}_{1-\alpha}(\cdot)$ is the paired-bootstrap $(1 - \alpha)$ lower confidence bound. The threshold is exactly zero: the definition admits no engineering-equivalence margin, and if one is wanted it has to be introduced explicitly and justified.

This is an empirical rule, not a $1 - \alpha$ confidence set, and we were wrong to call it one.

Each test is valid *for a fixed reference*. Ours is not fixed: $\hat{z}$ is the argmin of the same demonstrations the test then uses, so it is the most favourable candidate in this draw and candidates near it are rejected more often than α allows. Inverting a test does not handle that selection. The size of the effect is easy to see under a null of K candidates with identical risk, where every candidate is a true minimiser: at $K = 50$

and $M = 98$, matching our grid and demonstration count, a nominated true minimiser is retained about 53% of the time against a nominal 95% (`scripts/check_selection_bias.py`). Increasing the bootstrap count does not help, because the problem is the reference rather than the resampling. We therefore make **no coverage claim** for Eq. (2) as computed, and read it as what it is: a reproducible empirical rule for which settings this loss and this grid cannot separate from the best one.

A variant that removes the selection dependence, and what it says.

Selecting the reference on one half of the demonstrations and testing on the other makes selection and inference independent. On the same null that variant retains a true minimiser 92.4% of the time — most of the gap closed, with a few points of residual shortfall from the percentile bootstrap itself at $M/2 \approx 49$, so we call it a large improvement rather than an exact procedure. We report it alongside the same-sample rule wherever the distinction could matter, and Sect. 4.4 shows that the paper’s substantive conclusion — that the benchmark’s nominal setting is excluded — holds under it on both simulator stacks and across five random splits. Directions in which the loss is invariant (Sect. 4.2) enter C_α at every magnitude regardless of where the minimum lies, which is a stronger statement than “the data are consistent with a range of values”: along such an axis there is no range, the axis is unconstrained. Parameters the replay cannot observe at all — contact friction and object density never appear in a free-space trajectory — are likewise unconstrained, and we flag them separately because their compatibility is a property of the protocol’s coverage rather than of its precision.

Two objects follow from this and must not be conflated, because in our data they differ. C_α is the set Eq. (2) defines, computed against the loss minimiser over a stated search domain; Sect. 4.4 computes it and finds that it *excludes* the simulator’s nominal setting. F_{nom} , defined in Eq. (4), is the fibre of settings that leave the replay loss unchanged *relative to nominal* — the invisible axes through the point at which the benchmark actually runs. Our verdicts range over

F_{nom} , because every published rate is computed at nominal, and we say so wherever it matters.

3.3 The decision problem

Given a parameter set S , an evaluation budget and a policy pair, we must either declare an ordering or abstain. Two criteria are compared throughout:

- **point calibration**, standard practice: evaluate at a single setting and declare $i \succ j$ when the interval for $\hat{\Delta}_{ij}$ there excludes zero;
- **the union bound**: declare only when every $z \in S$ agrees in sign, using the interval $[\min_z \text{lo}(z), \max_z \text{hi}(z)]$.

Throughout the empirical sections $S = F_{\text{nom}}$ and the single setting is nominal, which is the pairing the benchmark’s own reporting implies. A third rung, a Gaussian-process simultaneous band over the parameter axes, is introduced in Sect. 8 for the case where the decisive parameter value may lie *between* the settings we sampled; the union bound’s guarantee is of sampling-extremum type and does not cover that case.

3.4 The error ledger

Writing $[L_{ij}, U_{ij}]$ for the union-bound interval, the claim that i is genuinely better in practice requires three events to hold simultaneously: that the true parameter lies in the set the bound ranges over, $E_\theta = \{z^* \in S\}$; that the evaluation intervals cover the simulated differences uniformly over the set, E_{MC} ; and that the sim-to-real bias is bounded, $E_b = \{|\Delta_{ij}^{\text{real}} - \Delta_{ij}^{\text{sim}}(z^*)| \leq b_{ij}\}$. On the intersection every real difference lies in $[L_{ij} - b_{ij}, U_{ij} + b_{ij}]$, and *if* the three carry levels $1 - \alpha$, $1 - \gamma$ and $1 - \beta$, a union bound gives $1 - \alpha - \beta - \gamma$ without requiring independence.

The conditional is doing real work and we leave it explicit. This paper establishes none of the three levels. E_θ would need a set with a coverage guarantee, and Sect. 3.2 explains why ours has none as computed; E_{MC} is a simultaneous statement and we report per-pair intervals; E_b needs a sim-to-real bias bound we do not have. The ledger is therefore an accounting of what a defensible claim would require, and a record of which parts we supply — not a derivation of a number attached to our results.

We are deliberate about which of these this paper establishes, and the accounting is less favourable than the arithmetic above suggests.

E_{MC} is what Sects. 5–7 measure, and enumerating $\mathcal{C}$ removes one of its two components exactly. Two caveats attach to it: our intervals are computed per policy pair, so what we report is a collection of per-pair statements rather than the simultaneous event E_{MC} written above, and we label them as such throughout rather than applying a multiplicity correction we have not performed.

E_θ is where the ledger is weakest, and there are two separate reasons, neither of which we repair. The first is Sect. 3.2: C_α as computed selects its reference on the same demonstrations it tests on, so it carries no $1 - \alpha$ guarantee either, and the split-sample variant retains a true minimiser 92.4% of the time against a nominal 95%. The second is Sect. 4.4: even a valid C_α would not transfer to F_{nom} , the fibre our verdicts actually range over, because F_{nom} is generated by a point the calibration data reject — nominal’s ratio and delay are identified and lose to the grid minimiser on both stacks. So for the sets we actually use we claim no coverage probability for E_θ at all. What we can say is narrower and still useful: F_{nom} contains every setting the calibration evidence cannot distinguish from the benchmark’s operating point, so a verdict that survives the union over it is one the calibration evidence cannot overturn *without also moving off the operating point*. That is a conditional statement, and we keep it conditional.

E_b we do **not** establish: we have no per-pair sim-to-real bias bound, and Sect. 8.4’s delimiting experiment is best understood as evidence that on at least one pair b_{ij} is large. A criterion built on a conditional E_θ and a per-pair E_{MC} bounds the ambiguity it can represent and no more, and that is the honest description of what the ladder in Sect. 8 delivers.

4 The calibration protocol is structurally blind

4.1 Setup

Calibration of the kind used by these suites fits simulator parameters by replaying recorded real demonstrations: the recorded end-effector targets

are fed to the simulated controller, and the parameters are chosen to minimise the deviation between the simulated and recorded trajectories. Let z collect the joint PD stiffness k , the damping d , the torque limit τ_{max} , the execution delay, and the contact parameters (object friction and density). The demonstrations are teleoperated, free-space and slow, which is the property that drives everything in this section: the protocol does not choose its own excitation.

The objective, written out, because its units matter later.

We inherit the reference implementation’s system-identification objective. For demonstration m of M , with T_m recorded steps, simulated end-effector position p_t against recorded $\hat{p}_t$ and orientation R_t against $\hat{R}_t$,

$$L_m(z) = \frac{1}{T_m} \sum_{t=1}^{T_m} (e_t^p + e_t^r), \quad (3)$$

where $e_t^p = \|p_t - \hat{p}_t\|_2$ is a position error in *metres*, $e_t^r = \arcsin(\|R_t - \hat{R}_t\|_F / (2\sqrt{2}))$ is an orientation error in *radians*, and $L(z) = M^{-1} \sum_m L_m(z)$. Adding the two makes this a **composite** objective: it adds a length to an angle, weighting one radian as one metre. That weight is inherited, not derived, and the sum is therefore not a distance, and in particular it must not be scaled by 1000 and reported in millimetres. Values of L and differences in L are quoted below in *composite units*, and wherever the physical size of an effect matters we give the translation and rotation components separately. For orientation: at the fitted setting $L = 0.0456$, of which the translation part is 19.1 mm and the remainder is the rotation term.

4.2 A structural insensitivity, not a flat region of a noisy objective

Two directions in z leave L unchanged to within magnitudes we quantify in Sect. 4.3. We are careful about the modality: each rests on a mechanism with a stated domain of validity, not on an exact identity of the full dynamics, and each is a property of the *protocol* rather than of the numerical optimiser — which is the distinction that matters, because no improvement in fitting removes it.

(i) Common scaling of the PD gains.

Under free-space replay at demonstrated velocities, the joint response to a target is dominated by a first-order lag with time constant d/k : the inertial, gravitational and Coriolis terms are small relative to the PD action at these speeds, and to that order scaling $(k, d) \mapsto (ck, cd)$ leaves the time constant, hence the trajectory, unchanged. *This is an approximation, and we state its limits.* Under the full rigid-body dynamics a common gain rescaling does not preserve the trajectory exactly — the inertial term does not scale with the gains — so the invariance is one of *degree*, holding while the neglected terms stay small compared with the calibration protocol’s own resolution. What makes it a structural rather than an incidental property is that the smallness follows from how the demonstrations were collected — teleoperated, slow, free-space — and not from a fortunate parameter value. Sect. 4.3 measures the residual: over $\times \frac{1}{4}$ to $\times 4$ it is at most 3.1×10^{-6} in the loss, against 1.36×10^{-3} between two implementations of the identical nominal dynamics. Outside that range, and at speeds that excite the inertial terms, we make no claim; a faster demonstration set would break it, which the remark below treats as the point rather than as a caveat.

(ii) The torque limit.

Wherever the commanded torque stays strictly below $\tau_{\max}$ the saturation is inactive and the trajectory does not depend on its value at all — not to a tolerance, but because the limit never enters the computation. This one is exact where it applies, and what bounds it is the domain rather than the mechanism: the argument holds for every demonstration that does not approach the limit, and says nothing about any that does. It therefore supports insensitivity over an *interval* of $\tau_{\max}$ — down to the largest commanded torque the demonstrations actually produce — and not down to arbitrarily small limits, which would eventually clip the demonstrated motion. We measured it at $\tau_{\max}/2$, where 97 of our 98 demonstrations fall in the inactive case (Sect. 4.3); the 98th is evidence that the interval’s edge is not far below that, and we claim nothing past the value we tested.

Both invariances are properties of the protocol’s *excitation range*, not of the simulator. A faster or more contact-rich demonstration set

would excite gravitational and Coriolis terms that break the first, and would drive the commanded torque into saturation, breaking the second. That is the point rather than a caveat: the blindness is inherited from demonstrations that were collected for teleoperation rather than for identification, and it would shrink under an excitation designed to identify. What cannot shrink it is more of the same data, or a better fit to it — which is the distinction between coverage and precision that Appendix A makes formal.

The ratio direction d/k and the execution delay *are* identified, and both produce the expected bowl-shaped loss curves; the claim is specific to the two directions above.

4.3 Empirical confirmation on two independent simulator stacks

We replayed 98 BridgeData V2 demonstrations (Walke et al., 2023) on (a) the ManiSkill3 port (Tao et al., 2024) and (b) the original reference stack (ManiSkill2 (Gu et al., 2023) with SAPIEN (Xiang et al., 2020)), the latter running headless through a purpose-built Vulkan layer (Appendix B).

All 14 iso-scale groups have maximum trajectory differences below 1 mm; we quote that threshold as a round descriptive figure, not as a tolerance anyone has defended. The paired *loss* increases for the ratio direction are $+0.0089 [+0.0073, +0.0105]$ and $+0.0064 [+0.0055, +0.0072]$ in composite units — three orders of magnitude above the 3.1×10^{-6} the common scale produces over a larger range, which is the contrast this section exists to establish. Figure 1 shows the bowl-shaped loss along the identified directions against the flat loss along the insensitive ones.

Three quantities appear in this section and they are not interchangeable, so we name them once. The 0.141 and 0.207 mm in Table 1 are maximum *end-effector position* differences between parameter settings, in metres. The loss increases just quoted are differences in the *composite* objective of Eq. (3), which is not a length. And neither is a residual of the fit: the fit’s own residual at the fitted setting is $L = 0.0456$ composite, of which 19.1 mm is translation. Sect. 4.4 judges parameter effects against a between-stack reference magnitude in the same composite units. Each

Table 1 Replay identifiability on two independent simulator stacks. 98 BridgeData V2 demonstrations. The two insensitive directions are flat to sub-millimetre and to bitwise on 97 of 98 demonstrations respectively; the identified directions produce the expected loss increase. The scale to judge these against is the 1.36×10^{-3} composite disagreement between these two stacks at *identical* nominal parameters (Sect. 4.4), whose translation part is 0.47 mm. Sect. 4.4 states what that magnitude is and is not: an engineering reference with an external basis, not a resolution limit below which an effect becomes unattributable.

Direction	ManiSkill3	Original stack	Verdict
Common scale $c \in \{0.25, 0.5, 2, 4\}$	max traj. diff. 0.141 mm	0.207 mm	invariant
Torque limit $\times 0.5$	bitwise 97/98, worst 0.149 μm	97/98, worst 0.050 mm	invisible
Ratio $d/k \times \{0.5, 2\}$	+0.0089 / +0.0064	same sign and order	identified
Execution delay 1 step	clearly identified	clearly identified	identified

is quoted in its own terms throughout and none is a substitute for another.

Halving the torque limit leaves the replayed trajectory *bitwise* identical on 97 of the 98 demonstrations, on both stacks independently, consistent with what Sect. 4.2(ii) predicts for a saturation that never activates. The exception is instructive and we report it rather than round it away: it is the same demonstration on both stacks, and there the trajectories differ by 0.149 μm on ManiSkill3 and 0.050 mm on the original stack.

We are deliberately cautious about why. That the same demonstration is the exception on both stacks is suggestive of a property of that demonstration rather than of per-stack numerical noise, but it is not decisive: two implementations of the same equations replaying the same trajectory can share a sensitivity, and the common cause could be the trajectory rather than the torque limit. The commanded torques were not recorded in the replay archive, so **we cannot confirm the saturation mechanism from the stored data and do not claim to**; it remains the natural explanation and an untested one, and re-running the replay with torques logged would settle it cheaply. What the data do support without an assumption is the magnitude: the trajectory is unchanged on 97 of 98 demonstrations and changes by 0.050 mm on the remaining one, which is a ninth of the 0.47 mm of translation inside the composite disagreement between the two stacks at identical parameters, and well inside any tolerance the fit could resolve (`scripts/audit_exact_invariance.py`).

4.4 The compatible set, and the fact that it excludes the benchmark’s own operating point

Inverting the calibration test requires a search domain, a loss, a reference and a threshold, and the answer depends on all four, so we state them. The domain is the 50-point grid of (stiffness scale, damping scale, execution delay) that the replay sweep enumerates; the loss is the per-demonstration composite objective of Eq. (3), which sums a position error in metres and an orientation error in radians; the reference is the grid’s loss minimiser; and a candidate is retained when the one-sided 95% lower bound of its paired mean loss increase over that minimiser does not exceed zero, which is the rule Appendix A states. Computed that way on the 98 common demonstrations (`scripts/rebuild_compatible_set.py`), the loss has a sharply two-part structure.

The loss depends on $(d/k, \text{delay})$ and on nothing else.

Grouping the 50 grid points by ratio and delay gives 26 groups, of which 14 contain two or more points that differ only in the common scale. Across those 14 — spanning seven distinct ratios from 0.5 to 2.0 — the mean loss moves by at most 7.2×10^{-6} within a group, identically on both stacks. *Between* groups it ranges over 6.0×10^{-2} , which is 8,354 times the largest within-group spread on ManiSkill3 and 8,767 times on the original stack. Extending to the sixteenfold range the policy sweeps actually use ($\times \frac{1}{4}$ to $\times 4$, measured on the original stack), the largest mean paired loss difference from nominal is 3.1×10^{-6} . All loss figures in this section are in the composite units of Eq. (3), not in metres.

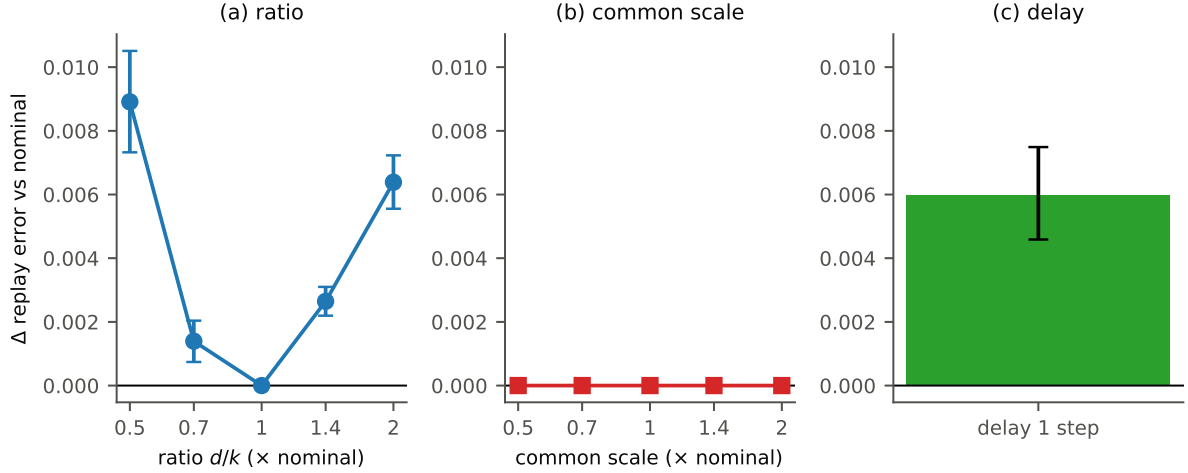


Fig. 1 Replay identifiability. Loss along the identified directions (the gain ratio d/k and the execution delay) is bowl-shaped with a well-defined minimum; loss along the invariant directions (common gain scaling, torque limit) is flat — for the torque limit, bitwise flat on 97 of the 98 demonstrations and within 0.050 mm on the remaining one. Panels share the vertical axis.

This is the insensitivity of Sect. 4.2(i) measured on the calibration objective itself rather than on a trajectory, and it is the sharpest form in which we can state it. Note what it is and is not: a property verified at seven ratios, not proved at all of them. Ratio 0.25 has exactly one grid point per delay value, so the common scale is not varied there at all — which matters in Sect. 7.7, because 0.25 is the ratio the calibration data prefer.

The threshold in Eq. (2) degenerates on a deterministic simulator.

Before reading the set off, the rule has to be examined, because at threshold zero it does something indefensible. Replaying demonstration m at parameter z is deterministic — the same loss every time — so the only randomness is which demonstrations were drawn, and with 98 paired demonstrations the bootstrap resolves mean differences of order 10^{-7} in composite loss. At threshold zero the rule therefore *rejects* $\times 2$ and $\times 4$ common rescalings whose mean loss differs from nominal by 4.9×10^{-7} and 7.4×10^{-7} — some two to three thousand times below τ — and as the demonstration count grows the retained set shrinks to the single loss minimiser whatever the physics. Statistical significance is the wrong question to ask of a difference that carries no noise; magnitude is the right one.

A reference magnitude whose basis is external to the conclusion.

Since the threshold cannot be zero, something has to set the scale, and we would rather it were not us. The two stacks implement the *same* nominal dynamics, and on the same 98 demonstrations their losses differ by 1.36×10^{-3} in composite units, 95% CI $[0.61, 2.10] \times 10^{-3}$, decomposing into 0.47 mm of translation and 0.88 mrad of rotation; the mean absolute per-demonstration disagreement is 2.76×10^{-3} . Two independent implementations of one set of equations differ by that much, so we use it as a reference magnitude $\tau = 1.36 \times 10^{-3}$, reading the set at τ as well as at zero and reporting 2.10×10^{-3} as a sensitivity.

We state its status carefully, because it is easy to overstate. This is an *engineering reference* with an external basis, not a resolution limit: a same-stack paired parameter effect can be both smaller than a fixed cross-stack offset and perfectly well resolved, since the two quantities have different sources of variation. What τ buys is a scale that was not chosen to produce an answer. It does not license the claim that anything below it is unattributable. Nor is the choice among the signed mean, the mean absolute difference and a quantile forced — they differ here by a factor of two — and we report which we used.

The insensitive directions survive every rule; the nominal setting does not clearly survive any.

Against τ , the iso-scale family ($\leq 3.1 \times 10^{-6}$) is retained by a factor of 440 and the halved torque limit ($\leq 1.8 \times 10^{-8}$) by a factor of 75,000. No conclusion in Sects. 4.2 and 4.3 depends on the threshold.

Nominal is the opposite case. The grid’s loss minimiser is not the simulator’s shipped controller: on *both* stacks independently it lies at one step of execution delay and a gain ratio of 0.25 to 0.36, while nominal sits at ratio 1 with no delay and ranks fifth of 26 groups. Nominal’s paired mean loss increase over the minimiser is $+2.90 \times 10^{-3}$ with a 95% lower bound of $+2.25 \times 10^{-3}$ on ManiSkill3, and $+2.26 \times 10^{-3}$ with $+1.49 \times 10^{-3}$ on the original stack; the ManiSkill3 figure decomposes into 1.06 mm of translation and 1.84 mrad of rotation. Both bounds exceed zero and both exceed τ ; at the 2.10×10^{-3} sensitivity nominal is still out on ManiSkill3 and comes in on the original stack. So its membership is **borderline**, and the exact grid point depends on the objective’s rotation weight: under weights from 0 to 5 radians per metre the ManiSkill3 minimiser is unchanged while the original-stack minimiser moves among three adjacent points. *What survives every weight is the qualitative statement* — both stacks prefer one step of delay and a ratio well below 1, and neither prefers nominal. We report that rather than a single privileged setting, because “the point the data identify” is only defined once the objective is.

The exclusion survives a procedure whose reference is selected independently.

The rule above selects its reference from the same demonstrations it tests on, which Sect. 3.2 shows invalidates the coverage reading, so the exclusion has to be checked against something better before any weight is put on it. Selecting the reference on one half of the 98 demonstrations and testing on the other gives independent selection and inference. Nominal is still excluded: on ManiSkill3 its paired mean loss increase over the half-sample reference is $+2.80 \times 10^{-3}$ composite with a one-sided lower bound of $+1.90 \times 10^{-3}$, and on the original stack $+2.49$ and $+1.43 \times 10^{-3}$. Across five random splits the lower bounds run from $+1.02$ to $+3.03 \times 10^{-3}$ and nominal is excluded in all ten

stack-split combinations, with the minimiser at delay 1 and ratio 0.25 or 0.35 every time. The conclusion of this subsection does not depend on the invalid step, which is why we keep it.

Two consequences follow, and we separate them carefully. Our earlier analysis fixed nominal *as* the test reference, which makes its own loss increase zero by construction and so could never reject it; the issue was invisible for that reason, not because the data spoke in its favour.

First, the invariance results of Sects. 4.2 and 4.3 are untouched and in fact strengthened, for the reason just given: they are statements about directions along which the loss does not move, they hold under every threshold we examined, and nothing about them depends on where the minimum sits.

Second, the object our verdicts range over is *not* C_α , and we stop calling it that. All of our policy sweeps are anchored at nominal: the conditions we evaluate are nominal together with the settings that leave the replay loss unchanged relative to it — common rescalings $\times 0.25$ and $\times 4$, the halved torque limit, and contact parameters the free-space replay never sees at all. That collection is the *calibration-invisible fibre through the benchmark’s operating point*,

$$\begin{aligned} F_{\text{nom}} = \{d/k = 1, \text{ delay } 0\} \times \{\text{scale} \in [\tfrac{1}{4}, 4]\} \\ \times \{\tau_{\text{max}}/\tau_{\text{max}}^{\text{nom}} \in [\tfrac{1}{2}, 1]\} \\ \times (\text{contact axes, unobserved}), \end{aligned} \quad (4)$$

where the scale and torque intervals are the ranges over which we *measured* the loss to be flat, not the ranges over which we assume it. This matters: the invariance argument of Sect. 4.2 is a statement about a mechanism, and a mechanism has a domain. A common gain rescaling leaves the free-space response unchanged to the order the demonstrations excite, which our measurements confirm over $\times \frac{1}{4}$ to $\times 4$ and which we do not claim beyond it; an inactive saturation limit is invisible only while it stays inactive, which bounds the torque statement below rather than at arbitrarily small τ_{max} . The contact axes are different in kind — unobserved rather than measured flat — and remain unbounded, which is a statement about the protocol’s coverage.

F_{nom} is the right object for the question we actually ask — *given that the benchmark runs at nominal, what does the calibration evidence determine about the numbers it publishes?* — because every published rate is computed at nominal. It is not the right object for the different question of what the calibration evidence determines about the simulator, and we do not use it for that. F_{nom} and C_α overlap only in the axes along which the loss is flat; they disagree on the ratio and the delay, which the data identify and on which nominal loses.

This raises a question we could evade but should not: if the benchmark’s operating point is one the calibration data reject, does the ranking computed there survive a move to the point they prefer? We answer it by measurement rather than argument. Sect. 7.7 re-runs the complete 64-configuration eggplant census at ($k \times 2$, $d \times 0.5$, delay 1), on the same configurations and the same policy seeds, so that the two operating points differ in nothing else. The answer splits by criterion. The point-calibration verdict does not survive: it moves from abstention to a declaration. The set-valued verdict does, abstaining at both points — which is what a criterion that reports the range of answers the evidence permits is for.

The question of this paper is therefore what a policy ranking computed over F_{nom} is worth, with the separate finding that the operating point generating that fibre is one the calibration data reject, and with Sect. 7.7 measuring what that costs.

5 What a finite benchmark measures

5.1 The benchmark is a finite population of configurations

Both widely used ports map the episode index onto a fixed grid of initial states,

$$\text{pos} = \lfloor (\text{episode_id} \bmod (n_{xy}n_{quat})) / n_{quat} \rfloor, \quad (5)$$

$$\text{quat} = \text{episode_id} \bmod n_{quat}, \quad (6)$$

so the number of distinct initial configurations is $n_{xy} \cdot n_{quat}$, and any episode index beyond that

repeats an earlier scene exactly. Object scale is drawn from the asset table, but every object has a single admissible scale, so it adds no variation.

Three consequences follow, and all three are visible in published practice.

1. **Beyond the grid, episodes are exact repeats.** A “96-episode” evaluation of the eggplant task on the ManiSkill3 port is 64 distinct scenes, 32 of which are visited twice.
2. **The two ports do not evaluate the same benchmark.** For the eggplant task they differ in the number of orientations (3 versus 8) and in the quaternion values themselves: the best matching pair has $|\langle q_1, q_2 \rangle| = 0.54$. Per-episode agreement between the ports is therefore meaningless for that task. Spoon and carrot share their grids exactly — and their cross-port per-episode agreement is 0.71–0.96, against chance level for eggplant. That contrast is the control for this diagnosis.
3. **One scene is worth 1/24 or 1/64.** On a 24-configuration task, any reported difference below 0.042 is less than one scene.

5.2 The official protocol is already a census

The reference evaluation script runs each bridge task over episode range 0 to 24 under three fixed policy seeds: the **complete 24-configuration census repeated under three fixed policy seeds**, 72 episodes, which is exactly the denominator of the published table (for instance $31/72 = 0.431$ and $41/72 = 0.569$ for the two Octo variants on the eggplant task). The ManiSkill3 port’s evaluation script instead defaults to 100 episodes indexed as seed + i , aligning with neither the grid nor the seed repetition. The community’s own reference protocol already does the right thing; the drift is in what downstream work does with it.

5.3 What counts as a replicate depends on the policy

A policy seed is a replication axis only if the policy consumes it. Octo’s diffusion action head draws a pseudorandom key that enters sampling, so changing the seed produces a new sample and configuration $\times$ seed is the legitimate replication axis. OpenVLA decodes greedily with sampling

Table 2 Distinct initial configurations per task in the two ports. The eggplant task differs in both the number of orientations and the quaternion values, so an episode index does not denote the same scene across ports.

Task	Original stack	ManiSkill3 port
Put eggplant in basket	$8 \times 3 = \mathbf{24}$	$8 \times 8 = \mathbf{64}$
Put spoon on towel	$12 \times 2 = 24$	$12 \times 2 = 24$
Put carrot on plate	$12 \times 2 = 24$	$12 \times 2 = 24$
Stack green on yellow cube	$24 \times 1 = 24$	$24 \times 1 = 24$

disabled, so the seed is unused and changing it produces nothing new; configuration is the only axis. Empirically, two “seed sets” of the deterministic policy are record-identical in 48 of 48 episodes. For such a policy, “replicated across seed sets” is vacuous, and the configuration grid — which can be enumerated — is the only sampling axis. This is the error behind the second retraction in Sect. 1.2.

5.4 Two estimands, two intervals

Two different quantities are both legitimate, and they carry different uncertainty.

(i) *The benchmark value* — the number this benchmark reports. After a census, configuration sampling error is zero and only policy noise and run-to-run numerical noise remain, so

$$\text{Var}(\hat{\Delta}) = N^{-2} \sum_c [s_A^2(c)/S_c + s_B^2(c)/R_c], \quad (7)$$

with S_c and R_c the number of runs of each policy at configuration c . “Run” means one episode, and the count is pooled across seed-set directories: a configuration that received six episodes contributes $S_c = 6$, even where those episodes came from fewer directories. That choice matters — an implementation that instead averaged within a directory and took each directory mean as the observation moves two verdicts — and Sect. 7.2 reports both alongside the reason for preferring this one.

(ii) *The generalisation value* — performance on comparable scenes. Here configuration sampling error is real and the appropriate estimator is a paired bootstrap *clustered by configuration*.

One term in Eq. (7) has to be imputed and we state the rule rather than leave it in the code. A configuration with a single run supplies no within-configuration variance, so $s^2(c)$ is unavailable there. Where the cell has other configurations with two or more runs we substitute the mean

of the measured s^2 ; where *no* configuration has a replicate we fall back to the binomial $\bar{p}(1 - \bar{p})$; and for a policy that decodes deterministically, whose re-runs are record-identical, we set the term to zero and label the cell as carrying no measurement of run-to-run drift. Of the 18 rows in the release’s core table, 15 are fully empirical on both sides, 2 use the deterministic-zero term for the OpenVLA side, and 2 use the binomial fallback. Every row prints its model, so a reader can see which interval rests on an imputation; the imputation is a pooled variance, which is an assumption of homogeneity across configurations that we do not test and that would be optimistic if run-to-run noise were concentrated in a few scenes.

Three further implementation points matter and are easy to get wrong. With a single run per configuration, estimator (i) degenerates to the ordinary binomial interval: **a census does not narrow an interval; replicates do**. Enumerate the grid first, which removes double counting, then add runs, which removes policy noise. Record-identical re-runs of a deterministic policy must be merged into one observation, or the variance is halved for free. And estimator (ii) must cluster by configuration; in our data this widens intervals by a median factor of 1.04 on the 64-configuration task and up to 1.33 on a 24-configuration task run for 96 episodes, and changes no verdict — but the effective sample size must be reported honestly.

5.5 The evaluation protocol and the implementation build are themselves invisible parameters

Holding the configuration grid fixed, we separated three further sources.

Remark 1 (Scope of the quantities in this paper). Unless a number is explicitly labelled otherwise, every effect size and interval we report is census-scope at five seed sets where five exist: the task’s

configuration grid is fully enumerated, so configuration sampling error is exactly zero, and the policy-noise term is estimated across as many independent seed sets as that task has. Five is the count for the eggplant census; the spoon and carrot sweeps have two, and Appendix C gives the per-cell figures. Where a quantity is also available at the three-seed budget the reference protocol uses, we give both and say which is which, because the difference between them is itself one of the paper’s results (Sect. 7.2).

The third row needs unpacking. `urdf_version` selects among four recolourings of the robot model. It changes no physical property — not a mass, not a friction coefficient, not a joint limit — and the published figure for this task is the mean over all four. On the fractal coke-can census the four variants give one policy 0.000, 0.280, 0.013 and 0.400, so evaluating on a single variant displaces the reported number by -0.173 to $+0.227$ relative to the four-variant mean. For comparison, the can’s physical orientation, which does change the dynamics, spans only 0.140.

This axis is invisible to replay calibration for the same reason the torque limit is (Sect. 4.2): it never enters the demonstration trajectory. Unlike the torque limit it is not even a physical parameter, which makes it the paper’s sharpest illustration that “calibration-invisible” is a property of the protocol’s coverage rather than of physics. That a visual variant should affect a vision-conditioned policy is not itself surprising — it is the premise of domain randomisation (Peng et al., 2018; Tobin et al., 2017).

Nor is the averaging itself undocumented: the suite describes its visual-matching design and reports the variant aggregation as part of the protocol (Li et al., 2025). Our contribution here is not the discovery of a hidden step but a statement about the *estimand* that step defines, and we want to state it precisely, because the loose version is wrong. The average over four variants is still a success rate — a perfectly well-defined one. What it is not is the rate *conditional on* a variant. The published number is the rate *marginal over* a uniform mixture of the four, and on this task the four conditionals span 0.400.

Two consequences follow, and neither is a variance claim. First, a reader comparing a published rate against a number obtained on one variant —

the natural thing to do when reproducing, since a single run fixes a single `urdf_version` — is comparing a marginal against a conditional. The discrepancy that results is -0.173 to $+0.227$ here with nothing having gone wrong in either computation. Second, and this is where it is easy to slip: under a complete enumeration the marginal carries *no* sampling error from the variant axis at all, because every variant is run. The 0.400 spread is therefore not uncertainty in the published number; it is the distance between that number and the four quantities a single-variant evaluation would produce. Treating the four as replicates of one quantity — and reporting a standard error over them — would misdescribe both. What we add is the magnitude and this bookkeeping, alongside the suite’s own observation that visual matching is what its simulated rates depend on.

Whether the axis can flip a *ranking* rather than widen one policy’s interval is a separate question, and Sect. 8.4 answers it on a second policy over the same grid. It can move a verdict to abstention, but it does not reverse an ordering: all twelve texture $\times$ condition cells there keep the same sign.

Both the seed and build rows are computed on matched episode scopes. That is not a formality: the directories differ in size — the pre-fix build and one seed set hold 96 episodes against 64 elsewhere — and because episode ids beyond the grid wrap onto it, averaging each directory over its own contents compares different effective scopes, which changes the answer by roughly a factor of two. We report the paired figures throughout and the unpaired ones nowhere, and `analyze_platform_drift_paired.py` prints both so the choice is inspectable. We do not, however, use the comparison to rank the build effect against policy-seed noise: a shift in Δ and a standard deviation of Δ are different quantities, and which is “larger” is not a well-posed question.

Two further practical consequences follow. First, one seed set plus a per-episode bootstrap declares rankings that do not hold: in our own data $\Delta(\text{small} - \text{base}) = +0.250 [+0.08, +0.32]$ declared one Octo variant better, and three seed sets from a single build give $+0.141 / -0.031 / -0.047$. We retract that verdict. Second, replicates must never be pooled across builds; doing so inflated the measured seed standard deviation from 0.055 to 0.093, so about half of the apparent seed noise was build drift.

Table 3 Sources of uncertainty that persist after the configuration grid is enumerated. The three magnitudes are *different kinds of quantity* — a standard deviation across seed sets, a paired mean shift between builds, and the spread of one policy’s rate across four settings — so the column reports each in its own terms and they should not be ranked against one another as though they shared a scale. Every entry is a quantity in units of Δ , the policy difference, and not a change in a single policy’s success rate — the two are easy to interchange and are not the same magnitude. The scopes differ per row. The seed row is census-scope at $S = 5$ on eggplant, median over policy pairs. The build row is a single run of each of two server builds on the eggplant census, so it carries no seed replication and no S ; it reproduces at the episode level on 0.73–0.88 of episodes. The texture row is one run per configuration over the 300-configuration fractal census, also without seed replication; the four variants give $\Delta = +0.213, +0.120, +0.200, +0.053$ there, while one policy’s *rate* over the same four spans 0.400 (0.000, 0.280, 0.013, 0.400) — a larger number that belongs to a different quantity and is quoted in the text, not here. Seed and build rows are paired on the episode ids the two directories share (`analyze_platform_drift_paired.py`).

Source	Isolation	Quantity	Magnitude
Policy seed	different seeds, same build	sd of Δ over seed sets	0.067
Implementation build	same seeds, newer build	shift in Δ	−0.109
Robot texture variant	<code>urdf_version</code> only	span of Δ over 4 variants	0.160

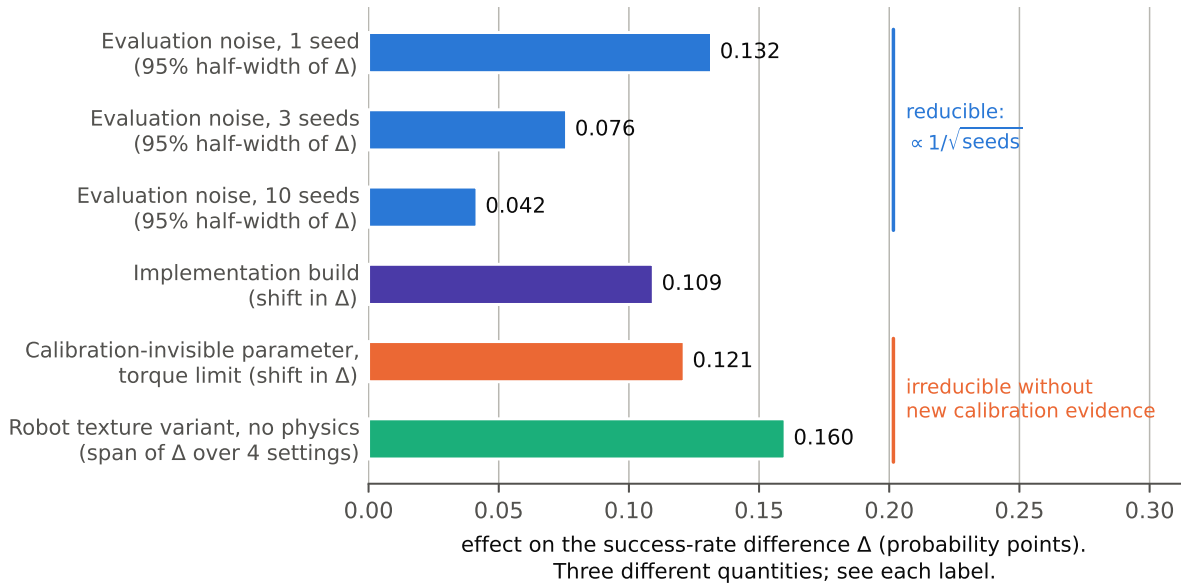


Fig. 2 The uncertainty budget after the configuration grid is enumerated. The point of the figure is which terms *respond to evaluation budget*: the three evaluation-noise bars shrink as $1/\sqrt{S}$, while the implementation build, the calibration-invisible torque limit and the robot texture variant do not shrink at all. Every bar is a quantity in units of Δ , the policy difference, but they are three different *kinds* of such quantity — a 95% half-width, a shift, and a span across four settings, as each label states. They share an axis because they share a unit, not because they are interchangeable, and their lengths should not be ranked against one another. Scopes differ per bar and are given in Table 3. Every value is computed from the episode records.

5.6 Reporting requirements

Every number in this paper carries: configurations covered / total; independent runs per configuration, and whether they are genuinely independent for that policy; the variance model (empirical, binomial fallback, or unmeasured); which estimand it targets; and the port and build it was collected on. We log the last of these per run in an

append-only record, having discovered that a per-directory metadata file is overwritten by the next run and makes the provenance of already-collected episodes unrecoverable.

6 Reproducing the published protocol

6.1 The protocol, followed exactly

The reference implementation’s bridge evaluation script evaluates each task over episode range 0 to 24 under three values of `init_rng`: a complete enumeration of the task’s 24 initial configurations, repeated three times, for the 72 episodes that form the denominator of the published rates.

How the seed is used inside a sweep is easy to get wrong, and we got it wrong first. The reference wrapper seeds *once*, in its constructor, and then lets one pseudo-random stream advance through every step of every episode of the run; its `reset()` restores the task, the image history, the action ensembler and the sticky-gripper state, and never touches the key. So a run’s 72 episodes carry 72 distinct noise realisations, and `init_rng` indexes a whole stream rather than a per-episode draw. Our first reproduction instead sent the seed on every reset, and our inference server re-seeded on receipt — with the consequence that all 24 configurations within a run replayed one identical noise realisation. We report both, because the difference is not cosmetic: it changes the agreement with the published table, the across-run spread, and one of our own conclusions (Sect. 6.3). All numbers below are from the reference lifecycle unless labelled otherwise.

We reproduced this on the original reference stack, running headless through the Vulkan compatibility layer of Appendix B, with the policies served by our own inference server rather than in-process. Published values are read from the pinned metrics table distributed with the benchmark. `scripts/queue_ms2_official_stream.py` runs the reference lifecycle and asserts, on every episode, that the server applied the lifecycle it asked for; `analyze_official_protocol.py --compare` prints the two side by side.

6.2 The reproduction agrees, and the differences are one-signed

Two things in Table 4 matter, and they pull in opposite directions.

The magnitudes are small. The policy-seed noise we measure on the 64-configuration eggplant census is ± 0.132 at half-width for a single seed set

and ± 0.076 for three (`analyze_seed_noise.py`), and every cell’s discrepancy is inside that band. Those two figures come from the spread of $\hat{\Delta}$ across whole seed sets — median sd = 0.068 over pairs and conditions, scaled as $1.96 \text{sd} / \sqrt{S}$ — which is a different noise model from Eq. (7), the estimator every table in this paper uses. Eq. (7) pools per-configuration run variances and gives per-pair three-seed half-widths of 0.075–0.088 on the same census (Sect. 7.6). The two answer different questions — how far a whole-census result moves between seed sets, against how precisely one census estimates the benchmark value — and we quote each where it belongs rather than treating them as one quantity. A different operating system, a software-rendered rasteriser, a different graphics path and an out-of-process policy server move the published numbers by less than one seed’s worth of noise, which is the external-validity check the rest of the paper rests on. On the second embodiment the same check gives +0.003 (octo-base on pick-coke-can, 0.173 against 0.170 published) and -0.040 (rt-1-x, 0.527 against 0.567).

But the differences are *one-signed*, and that is worth reporting even though we cannot turn it into a platform constant. Six of the eight cells fall below the published value, two match it, and none exceeds it; the mean is -0.038 . We resist three stronger readings that the numbers invite.

The eight cells are four tasks crossed with two policies that share seeds, so they are not eight independent measurements of one platform, and a t interval over them does not estimate a quantity that exists. The published rates are themselves point estimates without intervals, so part of the difference belongs to them. And the lifecycle comparison is suggestive, not causal: our earlier re-seeding run had a mean of -0.005 and a mean across-run range of 0.109 against 0.052 now, and while a noisier estimator agreeing more closely with a target is certainly not evidence of fidelity, three runs per cell is too few for those ranges to support a claim that extra variance *concealed* a real gap. What we can say is that the closer agreement we previously reported came from a protocol we now know to be wrong, and that the corrected protocol agrees less well and in one direction.

On magnitude, the arithmetic should be stated rather than rounded into a slogan. The mean absolute difference 0.038 is about a third of the 0.121

Table 4 Our 72-episode reproduction against the published values, under the reference RNG lifecycle. The rightmost column gives the same pooled rate from our earlier per-episode-re-seeding run, for comparison. Under the reference lifecycle the mean observed difference from the published table across these eight cells is -0.038 (mean absolute 0.038, largest 0.111), and six of the eight are below the published value with two at zero; under the re-seeding lifecycle the mean is -0.005 and the mean absolute 0.026. The eight cells are four tasks $\times$ two policies, sharing tasks, policies and seeds, so they are not eight independent replications and we quote the spread across them descriptively rather than as a confidence statement about a single platform parameter. Note also that the eggplant octo-base cell agrees to 0.4306 against a published 0.431, which is a rounding difference and not evidence of a downward shift.

Task	Policy	reference lifecycle				published	re-seeding	
		seed 0	seed 2	seed 4	pooled		diff.	pooled
eggplant	octo-small	0.542	0.500	0.542	0.528	0.569	-0.041	0.556
eggplant	octo-base	0.458	0.417	0.417	0.431	0.431	-0.000	0.472
spoon	octo-small	0.375	0.417	0.333	0.375	0.472	-0.097	0.431
spoon	octo-base	0.042	0.000	0.000	0.014	0.125	-0.111	0.083
carrot	octo-small	0.042	0.042	0.167	0.083	0.097	-0.014	0.069
carrot	octo-base	0.042	0.083	0.042	0.056	0.083	-0.027	0.125
stack cube	octo-small	0.000	0.042	0.042	0.028	0.042	-0.014	0.042
stack cube	octo-base	0.000	0.000	0.000	0.000	0.000	$+0.000$	0.000

torque shift, not an order of magnitude below it, and the largest single cell, 0.111, is close to it; against the 0.400 texture range the ratio is about ten. So the pipeline check licenses less than a sweeping claim of negligibility. What it does license is the specific use we make of it: every effect in Sects. 5–8 is a *difference between two policies measured in the same configuration on our own pipeline and one build*, where an offset common to both policies cancels and only its interaction with the policy pair would matter. The build condition is not decoration: Sect. 5.5 measures a build change moving $\hat{\Delta}$ by -0.109 , so it is not such an offset. The -0.038 is a caution about comparing our absolute rates with published absolute rates, which we do only in this section.

6.3 What the published budget resolves

The same table answers a second question, and the answer is less comfortable. Even under the reference lifecycle the three runs disagree with each other by more than we differ from the published value: the across-run range of a cell averages 0.052 and reaches 0.125. Applying the census estimator of Sect. 5.4 to the comparison the published protocol is built to support gives Table 5.

Three observations.

(i) One of four orderings is resolvable.

Only the spoon comparison excludes zero. On the other three tasks the published protocol reports a rate difference its own budget cannot distinguish from noise. This is not a criticism of the reference implementation — the published tables report rates, not rankings — but readers do compare those rates across policies, and on three of four tasks that comparison is unsupported.

(ii) A sign reversal we previously reported does not survive the corrected lifecycle, and we withdraw it.

Our earlier reproduction put octo-base above octo-small on carrot (-0.056) where the published values put octo-small above octo-base ($+0.014$), and we reported that as a concrete instance of a published ranking failing to reproduce. Under the reference RNG lifecycle carrot gives $+0.028$ — the same sign as published — and no cell in Table 5 disagrees in sign with the published table. The reversal was an artefact of our own per-episode re-seeding, which gave all 24 configurations in a run one shared noise realisation and so let a single unlucky stream decide the sign of a 0.014-sized difference. We therefore no longer claim that a faithful re-run of the published protocol reverses a published ordering; what we observed was our own implementation reversing it.

The underlying point survives in the weaker and better-supported form of observation (i):

Table 5 What the published 72-episode budget resolves, under the reference RNG lifecycle. Census interval: Eq. (7), policy noise only, $S = 3$ runs over a complete 24-configuration enumeration. The last two columns give our Δ and verdict from the per-episode-re-seeding run for comparison; the verdict column is unchanged by the lifecycle, the carrot sign is not. Generated by `scripts/make_table5.py`, which reproduces both.

Task	reference lifecycle			published Δ	re-seeding	
	our Δ	census 95%	verdict at this budget		Δ	verdict
eggplant	+0.097	$[-0.042, +0.236]$	abstain	+0.138	+0.083	abstain
spoon	+0.361	$[+0.259, +0.463]$	octo-small $\succ$ octo-base	+0.347	+0.347	octo-small $\succ$
carrot	+0.028	$[-0.039, +0.094]$	abstain	+0.014	-0.056	abstain
stack	+0.028	$[-0.011, +0.066]$	abstain	+0.042	+0.042	abstain

carrot’s difference is +0.028 with interval $[-0.039, +0.094]$, so the published +0.014 is a number whose sign its own budget does not determine. A comparison that cannot be resolved can come out either way, and in our first run it did — which is a fact about the budget, demonstrated on our own data rather than charged against the reference implementation.

(iii) Removing configuration sampling error is not enough.

The census interval is narrower than the binomial-over-72 a reader of the published table would compute on three of the four tasks and marginally wider on the fourth: half-widths 0.139 against 0.163 on eggplant, 0.102 against 0.115 on spoon, 0.067 against 0.083 on carrot, and 0.039 against 0.038 on stack. No verdict changes either way. With three runs per configuration the per-configuration policy variance dominates, so enumerating the configuration grid, which sets the sampling error to exactly zero, buys much less than practitioners expect. Budget is spent better on seeds than on episodes.

(iv) The verdict count depends on the variance model, and we say which we use and why.

“One of four” is not model-free. Eq. (7) treats each configuration’s runs as the replication unit and pools the per-configuration policy variances; on the new records that gives the intervals in Table 5 and one resolvable ordering. An alternative that treats each of the three runs as the unit — a paired t interval on three whole-census block differences — gives eggplant $[+0.038, +0.157]$, spoon $[+0.242, +0.481]$, carrot $[-0.188, +0.243]$, stack $[-0.032, +0.088]$, hence *two* resolvable orderings.

The block interval is not automatically better: with three blocks it rests on two degrees of freedom, and eggplant’s three block differences happen to fall close together, which is what makes that interval narrow.

We report Eq. (7) as the headline because it matches the randomisation the protocol actually performs. Under the reference lifecycle each episode draws from its own point in one advancing stream, so the noise varies within a run and not only between runs, and a model whose replication unit is the whole run attributes within-run variation to nothing. Under our earlier re-seeding lifecycle the opposite was true and the block model would have been the right choice — which is a further consequence of the lifecycle error, and the reason we state the model explicitly instead of inheriting it. The sensitivity is reported here rather than buried because the headline claim is one of the paper’s three, and a reader is entitled to know that a defensible alternative model makes it two of four rather than one.

6.4 What this section licenses

The published numbers carry an uncertainty of roughly ± 0.08 at three seeds, which the published tables do not quantify, of the same order as the parameter-induced shift of 0.121 that Sect. 7 attributes to a calibration-invisible torque limit. That these two are comparable is the paper’s central quantitative claim, and Sect. 6.2 establishes that we can say it about the benchmark as published, not merely about our port of it.

Two qualifications travel with that licence. Our rates average 0.038 below the published ones, so a comparison of an absolute rate of ours against a published absolute rate should carry that offset; every claim in Sects. 7 and 8 is instead a *difference*

between two policies measured in the same configuration and on one build, where an offset common to both policies cancels — a build change is not one of those (Sect. 5.5). And the RNG lifecycle is part of a protocol, not an implementation detail beneath it: it changed our agreement with the published table, halved the across-run spread, and decided the sign of one comparison. A reproduction that matches an outer loop while differing in how the policy’s randomness is threaded through it is not a reproduction of the same estimand, which is a small instance of this paper’s general point applied to our own work.

7 Decision-relevant non-identifiability

Sections 4 and 5 showed that the calibration protocol cannot see certain parameter directions and that the benchmark’s own budget leaves its rankings unresolved. Neither observation is interesting on its own: an unidentifiable parameter matters only if moving it inside the compatible set changes an answer someone acts on. This section establishes that it can, measures how large the effect is, and reports how much of it survives when the evaluation budget is increased — which is less than we first claimed.

7.1 The claim, stated so it can fail

Fix a task, a policy pair (i, j) and the invisible fibre F_{nom} of Sect. 4.4. The claim under test is that point calibration and the union bound disagree — that point calibration declares orderings the calibration data cannot support. Two forms are worth separating.

The **strong form**: some $z \in F_{\text{nom}}$ reverses a ranking’s sign, so that the simulator’s answer to which policy is better depends on a parameter the calibration data is silent about. **We report this as false.** Across every pair, task and condition we measured, no calibration-invisible parameter reverses a sign-determined ranking. We also once believed we had found a weaker version of it — a between-condition sign change on one pair — and withdrew that too (Sect. 7.5).

The **weak form**: the parameter moves Δ far enough to cross the decision boundary, turning a declared ordering into an abstention. This holds,

and it is budget-dependent — which is the finding that replaced our original, stronger claim.

7.2 The core table, and what a larger budget did to it

The complete per-pair table — both verdicts for every policy pair on every task, with the configurations covered, the runs per policy and the variance model for each — is `results/CORE.TABLE.md` in the release, generated by `make_core_table.py`. It has 18 rows: 17 bridge policy pairs across the two ports, plus the pick-coke-can pair of Sect. 8.4, which we count separately because it is a different embodiment and a different question. Wherever this paper says “of 17 pairs” it means the 17 bridge rows. Table 6 below is not that table; it shows only the three pairs whose point-versus-set verdict differs at the reference protocol’s three seed sets, and what raising the budget to five does to each.

Before the numbers, the observation unit, because the estimator has two defensible ones and they disagree on two rows. Eq. (7) writes S_c for the number of *runs* at configuration c , and that is what we report: episode outcomes pooled across seed-set directories, so a configuration that received six episodes contributes six observations. An earlier implementation instead averaged a directory’s episodes at a configuration and treated each *directory mean* as the observation, making S_c the number of directories. The two differ because seed set B carries 96 episodes on a 64-configuration grid, so ids 64–95 wrap onto configurations 0–31 and give those two episodes — at different policy seeds, since `policy_seed = base + episode_id`. Neither unit is wrong: the run unit is finer and is what the formula states; the directory unit is conservative against a directory-level common effect, which the build-drift result of Sect. 5.5 shows can exist across generations, though the five sets here share one build. We adopt the run unit because it is the formula’s, and report the other as a sensitivity — not the reverse, and not chosen by which declares more. `make_core_table.py --unit directory` reproduces it. The two units move two of the eighteen rows: `octo-small` vs `octo-base` on eggplant declares under the directory unit on a lower bound of $+0.000042$ and abstains under the run unit at $[-0.008, +0.117]$,

and `octo-small` vs `octo-base` on the original-stack spoon does the opposite. **The point-versus-set disagreement count is 1 of 17 under both.**

Of the 17 bridge pairs, **one** disagrees: `octo-base` vs `octo-base@hist1` on the eggplant census, point $\Delta = +0.073 [+0.01, +0.14]$ declaring `octo-base` better, union interval $[-0.06, +0.14]$ abstaining.

The union runs over the nominal setting as well as the five calibration-invisible ones, and the reason is definitional rather than statistical. The set being unioned over is F_{nom} , the fibre of settings the replay loss cannot distinguish from nominal (Sect. 4.4), and nominal is its base point, so it belongs to the fibre by construction. An envelope taken over the perturbed conditions alone would not cover the fibre it claims to bound. It widens no verdict in Table 6. We note that nominal’s membership in F_{nom} is a tautology and carries no implication that nominal is the calibration optimum — Sect. 4.4 shows it is not.

One consequence should be stated plainly, because a reader scanning the table’s two-decimal columns cannot see it. Two declarations rest on a bound that rounds to $+0.01$: `octo-small` vs `octo-small@hist1` at $+0.0091$, and the surviving disagreement’s own point calibration at $+0.0095$. Put a lower bound of $+0.0095$ in units of the evidence behind it. On a 64-configuration census at five runs per policy, $\hat{\Delta}$ weights each configuration mean by $1/64$ and each *individual episode* by $1/(64 \times 5) = 0.0031$, so $+0.0095$ is about 0.6 of one configuration’s mean outcome, or about **three single-policy episodes** out of the 320 on each side of the comparison. Three episodes flipping the other way would erase it. Neither is a declaration in any useful sense; both sit on the knife edge, and we flag them rather than let the rounding carry them. The criterion’s substantive declarations are the ones with margin: spoon $[+0.04, +0.56]$, the OpenVLA comparisons, and the fractal pair of Sect. 8.4. The generator marks every such row, so the distinction cannot be lost in a later revision. We also report, rather than bury, that the knife edges are not stable across the two observation units — the largest of them under the directory unit was $+0.000042$, on a pair that abstains under the run unit. A bound that small is a property of the estimator’s last digit, not of the policies.

At three seed sets, three rows disagreed. The two that no longer do were resolved by spending seeds, not by any change of method or of data-retention rule.

We had pre-committed to this test and to withdrawing a row if spending seeds resolved it, which is why the surviving count is one and not three.

The honest general statement is therefore about the evaluation budget. At the reference protocol’s three seed sets, point calibration declares orderings the union bound does not support on **3 of the 17** bridge policy pairs — though one of the three, as the caption says, abstains by six millionths. At five seed sets over a complete configuration census, one remains. A single seed set can evaluate only 13 of the 17 pairs at all, and disagrees on 2 of those 13. The criterion’s value is concentrated in the budget regime practitioners are actually in. This is weaker than the claim we set out to make, and it remains useful: we are aware of no published evaluation that runs five seed sets over a full configuration census, and the reference protocol itself uses three. Every count here is reproducible with `make_core_table.py --max-sets`, which is the option we added when we found that the historical counts could not be recomputed.

The surviving pair is also instructive about *why* it survives. It is not that its Δ changes sign across conditions (Sect. 7.5), but that its nominal interval is barely decidable (lower bound $+0.0095$) while the union bound takes the minimum lower bound over conditions, which friction $\times 0.4$ supplies at -0.064 . It is a margin-and-width phenomenon, not a parameter-sign phenomenon — and one whose point declaration sits on the knife edge, so the disagreement is between an abstention and a barely-made declaration rather than between two confident answers.

7.3 The mechanism: one parameter, measured on a complete census

The torque limit is the cleanest case. It is invisible to calibration (Sect. 4.2(ii)) because along demonstrated velocities the commanded torque does not saturate, so replay loss does not depend on it: bitwise on 97 of the 98 demonstrations, and to 0.050 mm on the one that evidently comes closest to the limit. Halving it therefore stays inside F_{nom} with a margin set by the mechanism

Table 6 The effect of raising the seed budget from three to five on the point-versus-set disagreements of the eggplant census. Both columns are recomputed on the run observation unit with $S = 3$ defined as seed sets A', B and C, so the two are directly comparable and both are reproducible from the release with `make_core_table.py --max-sets 3`. The third row's $S = 3$ abstention rests on a lower bound of -6×10^{-6} , which is a disagreement in arithmetic and not in substance; we list it because the count is ours to report, not to curate. Envelopes include the nominal setting, as the bound requires.

Pair	Union at $S = 3$	at $S = 5$	Outcome
octo-small vs octo-base@hist1	$[-0.020, +0.275]$ abstain	$[+0.012, +0.245]$ declare	withdrawn
octo-base@hist1 vs openvla	$[-0.000006, +0.416]$ abstain	$[+0.017, +0.426]$ declare	withdrawn
octo-base vs octo-base@hist1	$[-0.099, +0.167]$ abstain	$[-0.064, +0.137]$ abstain	survives

rather than by a fitting tolerance. On the 64-configuration eggplant census it moves the cross-family comparison by $-0.104, -0.146, -0.108$ and -0.127 for the four Octo-versus-OpenVLA pairs: a shift of $0.104-0.146$, mean **0.121**, uniform enough across pairs to be consistent with a single policy moving rather than four independent effects (`scripts/analyze_torque_shift_s5.py`).

Remark 2. Two earlier measurements of this shift were larger, and both were superseded for the same kind of reason. On 48 of the 64 configurations it read $0.23-0.25$, with two pairs crossing the decision boundary: the partial census overstated it. On the complete census at three seed sets it read $0.133-0.161$, mean 0.141 ; at five it reads as above. Every effect size in this paper is census-scope at $S = 5$ for this reason, which is the budget Sect. 7.2 establishes as the one that settles verdicts.

7.4 Why it is not an artefact

Three checks.

It is a property of one policy, and a large one.

Halving the torque limit raises OpenVLA's eggplant benchmark value from 0.156 to 0.271 on the complete 64-configuration census ($+0.115$), while octo-small, run as a negative control over the same census, moves by $+0.011$ and the other three Octo policies by -0.06 to $+0.02$ with no consistent direction. The asymmetry has a mechanism: OpenVLA emits larger single-step displacements, so at the nominal limit it pushes the eggplant away or re-grasps repeatedly on contact, and the episode-level counters agree (grasp rate $+0.12/+0.18$, fewer repeated gripper closures).

It is broad, not driven by a few configurations.

Over the complete census 16 configurations improve, 42 are unchanged and 6 degrade, and the mean change is positive in 7 of the 8 eggplant orientations (-0.04 to $+0.29$). The gain is anticorrelated with nominal success ($r = -0.44$): of the 51 configurations where OpenVLA fails outright at the nominal limit, 13 are rescued. The parameter repairs failures rather than padding successes. The same decomposition for octo-small is flat and directionless — 15 improve, 15 degrade, 2 of 8 orientations positive — which is what a policy that never saturates the limit should look like.

It is task-dependent in the direction the mechanism predicts.

On the spoon task (24-configuration complete census) the same parameter barely moves Δ at all — -0.01 to -0.07 across the four cross-family pairs, against -0.14 on eggplant — and changes no verdict. Eggplant is where contact torque saturates for a large-displacement policy; spoon is not. A calibration-invisible parameter is therefore not uniformly dangerous: its decision-relevance is a property of the task-policy pair, which is precisely why it cannot be argued away a priori and has to be enumerated.

7.5 A withdrawn claim: there is no between-condition sign reversal

An earlier version of this section reported that one pair, octo-base vs octo-base@hist1, reversed sign across the invisible fibre — $\Delta = -0.021$ under density $\times 0.5$ against positive values under the other five conditions — and argued from it that this pair's abstention was a fact about the parameters rather than about our budget, and so could never be removed by more evaluation.

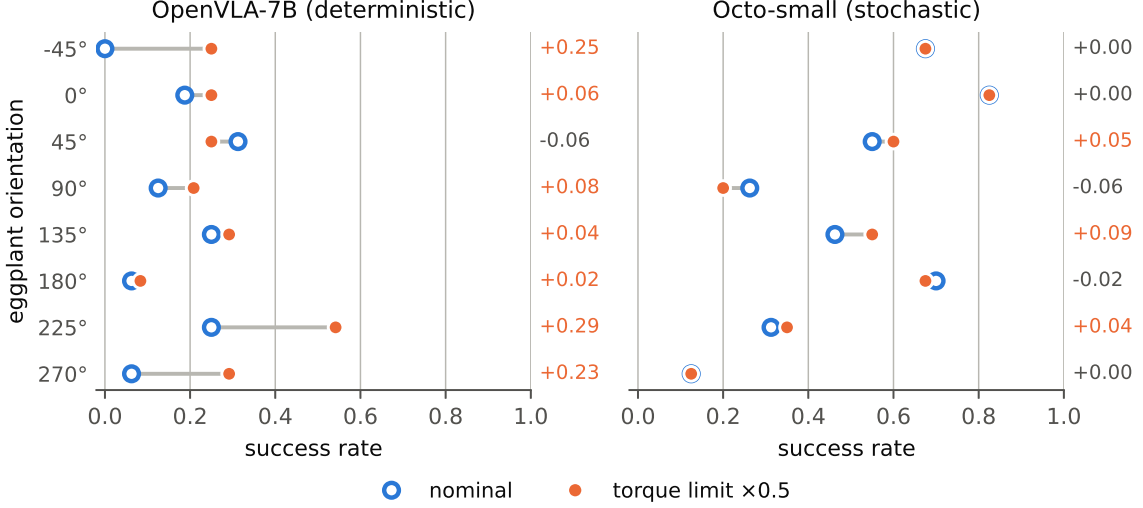


Fig. 3 The torque-limit effect decomposed over the eggplant orientation grid, for a deterministically decoding policy (OpenVLA, left) and a stochastic one (Octo-small, right). The effect is positive in 7 of 8 orientations for the former and directionless for the latter, which is the asymmetry the saturation mechanism predicts.

We withdraw that. At five seed sets the same cell reads $+0.042 [-0.022, +0.106]$, and no condition has a point estimate below -0.001 : nominal $+0.069$, force $\times 0.5 +0.050$, iso $\times 0.25 +0.030$, iso $\times 4.0 +0.006$, friction $\times 0.4 +0.000$, density $\times 0.5 +0.042$. The between-condition sign reversal we had reported is not merely smaller at five seed sets; it is absent.

The distinction the withdrawn claim was reaching for is nevertheless real and worth keeping, because Sect. 8 needs it. As the evaluation half-width $h \rightarrow 0$ the union bound converges to $[\min_c \Delta_c, \max_c \Delta_c]$, which does **not** shrink with budget. An abstention that survives $h \rightarrow 0$ is a parameter fact; one that dissolves as h falls was a power artefact. Our own three-to-five-seed experiment is the clean demonstration of the second case, and Sect. 8.4 uses the same test to reject an abstention we would otherwise have been tempted to claim as a success.

7.6 What determines whether a ranking can be declared

A ranking can be declared when $|\hat{\Delta}|$ exceeds the parameter-induced shift δ **plus** the evaluation half-width h . Enumerating the configuration grid drives the sampling part of h to zero; more seeds shrink the rest, as $1/\sqrt{S}$ — from 0.075–0.088 at three seed sets to 0.058–0.068 at five. The

parameter-induced shift $\delta \approx 0.121$ does not move, by construction.

The two regimes this produces are the practical content of the section. At the budget the reference protocol uses — 24 configurations, three seeds — a pair’s half-width is 0.05–0.13 (Sect. 6.3), which is at or above δ ; there the two error sources are not even separable, and a practitioner cannot tell an identifiability problem from a power problem. Our census at $S = 5$ pushes h below δ , and only then does the comparison become meaningful. The recommendation follows directly: report the estimand and the budget, and spend budget on seeds before episodes, because only then does the residual ambiguity that *cannot* be bought down become visible.

7.7 The ranking at the calibration-preferred operating point

Everything above is computed at the simulator’s shipped nominal controller setting, which Sect. 4.4 showed the calibration data reject in favour of $(k \times 2, d \times 0.5, \text{delay } 1)$. That leaves an obvious question, and rather than argue about it we measured it: we re-ran the complete 64-configuration eggplant census for `octo-small` and `octo-base` at the fitted setting, under three seed sets whose

bases match three of the census sets used throughout. The comparison is paired in everything that can be held fixed — the same 64 configurations, the same policy seeds episode for episode, the same policies, the same build, the same estimator (Eq. (7)) — so the only difference between the two columns of Table 7 is where the controller was set. Nominal is restricted to the same three seed sets, because comparing against the $S = 5$ interval would confound the operating point with the budget.

The verdict changes; the underlying difference is not resolved.

This distinction is the whole content of the result and we lead with it. What we observe is a *verdict* change on this batch: at nominal the interval includes zero and the criterion abstains, at the fitted point it excludes zero and declares. What we do *not* observe is a resolved change in the policy difference. Both $\hat{\Delta}$ are positive — there is no sign reversal — and the paired change itself, $D = \hat{\Delta}_{\text{fitted}} - \hat{\Delta}_{\text{nominal}} = +0.031$, is not distinguishable from zero under any sampling model we can defend.

The primary one has to match this paper’s own estimand, and that settles it: the 64 configurations are enumerated and fixed, so they are not a sampling unit, and what remains random is the policy seed. Each seed set supplies all four arms — both policies at both operating points — at every configuration, so it yields one complete paired shift $d_c^{(k)} = [\hat{p}_A^{\text{fit}} - \hat{p}_B^{\text{fit}}] - [\hat{p}_A^{\text{nom}} - \hat{p}_B^{\text{nom}}]$ there, and $\text{Var}(D) = N^{-2} \sum_c \text{Var}_k(d_c^{(k)})/K$ keeps the pairing rather than assuming the arms independent. That gives $D \in [-0.086, +0.148]$. Three sensitivities bracket it: summing the four arms’ run variances separately, which discards the pairing, gives $[-0.093, +0.156]$; treating configurations as a sampling unit — a superpopulation reading, and the estimand Sect. 5 argues is not the benchmark’s — gives $[-0.064, +0.126]$; whole seed sets as blocks on two degrees of freedom give $[-0.300, +0.363]$. The three seed sets individually give $+0.172$, $+0.016$ and -0.094 — one of the three moves the other way. All four intervals contain zero and none is an exact finite-sample guarantee. So a significant interval at one operating point and a non-significant one at the other

does not make the difference between them significant, and we do not claim a stable population effect.

What the measurement does establish is fragility, and that is the paper’s thesis rather than a weaker version of it. A change of 0.031 — less than half the ± 0.088 half-width at this budget, a quarter of the 0.121 torque shift, and itself unresolved — moves the answer from abstention to a declaration, because at nominal the lower bound already sat at -0.016 . The declaration at the fitted point rests on a lower bound of $+0.016$. This comparison is matched at three seed sets, so each policy contributes $64 \times 3 = 192$ episodes and each weighs 0.0052 in $\hat{\Delta}$: the bound is about **three single-policy episodes**, or one configuration’s mean outcome out of 64. Three episodes are what separates the declaration from an abstention. So at the budgets in use the verdict here turns on the operating point — and, unlike the modelling choices of Sect. 7.2, this is a choice the calibration evidence *does* settle, against the setting the benchmark uses. That is the C3(ii) case rather than the C3(i) case, and we keep them apart: what is unsettled is not which point the evidence prefers but which verdict is correct, since the difference between the two points is not itself resolved. A reader given only the published rates cannot see any of this, because those rates exist at one operating point and carry no interval.

The set-valued verdict does not change: it abstains at both operating points.

Taking the union over each point’s own calibration-invisible fibre gives $[-0.035, +0.300]$ at nominal, over its complete six-condition fibre, and $[-0.029, +0.212]$ at the fitted point, over the four of six planned conditions that are complete. Both abstain. The fitted figure is already final in the only sense that matters here: an envelope is a minimum of lower bounds and a maximum of upper bounds, so adding a condition can only push the bound outward, and a bound that already contains zero will still contain zero once the remaining two conditions are in. That is a monotonicity argument rather than a prediction about data we have not collected.

This corrects an earlier draft of this section, and the correction is worth recording because it illustrates the trap. Over the first two

Table 7 The same policy pair, the same 64 configurations and the same policy seeds, evaluated at the benchmark’s operating point and at the one its replay objective prefers. **The point verdict changes and the set verdict does not.** The replay-loss column is the composite objective of Sect. 4.1, in its own units, not a distance — see Sect. 4.4. The set envelope at nominal is over its complete six-condition fibre; at the fitted point four of the six planned conditions are complete and already abstain, and because a min/max envelope only widens as conditions are added, the remaining two cannot restore a declaration. Generated by `scripts/analyze_fitted_point.py`, which refuses any condition that is not a complete census.

Operating point	Replay loss		verdict	
	vs minimiser	$\hat{\Delta}$	point 95%	set envelope
nominal, $d/k = 1$, no delay	+0.0029 (rejected)	+0.073	$[-0.016, +0.161]$ abstain	$[-0.035, +0.300]$ abstain (6 cond.)
fitted, $d/k = 0.25$, delay 1	0 (the minimiser)	+0.104	$[+0.016, +0.192]$ small $\succ$ base	$[-0.029, +0.212]$ abstain (4 of 6)

completed conditions the fitted envelope was $[+0.009, +0.192]$ and declared, and we wrote that both verdicts flip. The third condition — a common rescaling of the fitted gains — gives $\hat{\Delta} = +0.057$ with $[-0.029, +0.144]$, which pulls the envelope across zero on its own. An envelope reported before its design is complete is not a conservative version of the final answer; it is a different and narrower one, and here it pointed the opposite way.

Both fibre memberships are measured rather than assumed, which took two additional replay experiments. The common-scale directions: the objective is flat along them in 14 of the replay grid’s 26 (ratio, delay) groups, across seven ratios from 0.5 to 2.0, to within 7.2×10^{-6} in composite units, but ratio 0.25 has exactly one grid point per delay value, so we swept the common scale sixteenfold at ratio 0.25 directly and found it flat to 5.4×10^{-6} on ManiSkill3 and 1.1×10^{-5} on the original stack. The torque limit: the argument at nominal is that the commanded torque never reaches the limit, and that does *not* transfer for free, because commanded torque depends on the gains and the fitted point doubles the stiffness. Measured at the fitted point, halving the limit leaves the replayed trajectory bitwise unchanged on 96 of 98 demonstrations against 97 of 98 at nominal, and on the demonstrations that do change the largest end-effector position difference is $84.5 \mu\text{m}$ against $50.1 \mu\text{m}$ — so the clip does activate a little more often and a little harder, as the mechanism predicts, while remaining far below the protocol’s own resolution. Both directions are retained at the fitted point on evidence rather than by inheritance. What the records cannot

show is the mechanism itself: commanded torques were not logged, so “the clip activates more often” remains the natural reading of these counts rather than something we measured directly.

One further observation, which corrects something we had stated too strongly. The union bound’s abstention at both operating points should not be read as the criterion *protecting* against the operating-point choice. An envelope over a set cannot speak about a point the set excludes, and nominal’s fibre excludes the fitted point by construction — the two differ in the ratio and the delay, which the calibration data constrain, so neither invisible fibre contains the other’s base point. A set containing both operating points and the region between them *would* have an envelope covering both; it would simply be much wider. So there is no third irreducible kind of ambiguity here, and we withdraw that framing: it is a question of set definition, and it identifies a choice the paper had left implicit. The invisible fibre is the right object for asking what the published numbers determine, and the wrong object for asking whether the operating point itself was well chosen. What addresses the latter is not a wider envelope but reporting the operating point and its standing against the calibration objective — which costs nothing, and which no benchmark we know of does.

8 A verdict ladder and its reach

Sections 4 through 7 establish that a simulation benchmark’s ranking claims are underdetermined

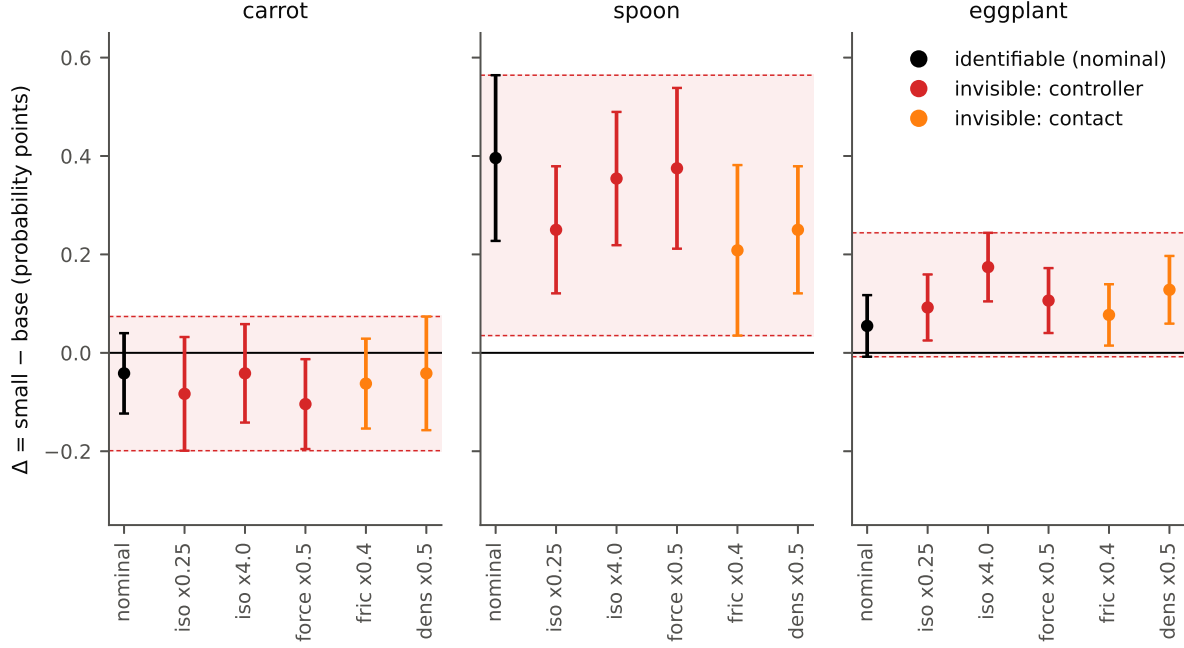


Fig. 4 Δ between two Octo variants by simulator condition, on three bridge tasks. Grey points are conditions the calibration data identifies, coloured points are the calibration-invisible ones, and the shaded band is the union bound over the invisible fibre — the invisible conditions *and* nominal, which the set contains by construction. Computed with the same estimator and the same seed sets as Table 6, so figure and table report one computation rather than two, on the run observation unit of Sect. 7.2. Verdicts and set bounds: carrot, point and set both abstain, $[-0.20, +0.07]$; spoon, both declare octo-small better, $[+0.04, +0.56]$; eggplant, both abstain, $[-0.008, +0.244]$. Available seed sets differ by task — five for eggplant, two for spoon and carrot — and the per-cell configuration and run counts are in Appendix C.

by the evidence behind them, from three independent directions. This section asks what to do about it: what a set-valued criterion buys, at what cost in power, and where it stops working. We answer all three, and the third answer is negative.

8.1 Two rungs, with different guarantees

The **union bound** declares an ordering when every *sampled* parameter setting in F_{nom} agrees in sign. Its guarantee is of **sampling-extremum** type: valid with respect to the extremes of $\Delta(z)$ over the settings actually run, and silent about values between them.

A **Gaussian-process simultaneous band** fits $\Delta(z)$ over the parameter axes and requires the sign to hold across a simultaneous band over the whole compatible region. Its guarantee is of **within-band** type under a smoothness prior, and it therefore covers features that fall between sampled settings — at a cost in power that Sect. 8.2 quantifies.

An earlier design had a third element, a flatness gate that chose between the two per axis. We removed it; Sect. 8.2 explains why.

8.2 Coverage and power on synthetic ground truth

Because real data provide no ground truth about $\Delta(z)$, we built a synthetic track in which the response surface is known and the criteria can be scored directly: 300 repetitions per cell over four surface families, crossed with true margins Δ_0 and episode budgets n .

(i) Point calibration’s error rate rises with the episode budget.

On any non-flat surface the nominal interval covers $\Delta(\hat{z})$ rather than the extremes over C_α , so more episodes shrink the interval around a value that was never the quantity of interest. On the linear surface at $\Delta_0 = 0.1$ the error rate goes from 0.13 at $n = 24$ to 0.36 at $n = 96$; on the sampled-dip surface at $\Delta_0 = 0.2$, from 0.26 to 0.88. This is

Table 8 Minimum coverage / maximum error rate / mean declaration rate on cells where the truth is decidable, by response-surface family. The union bound fails exactly where the decisive feature falls between sampled settings. All three columns are computed on one configuration — the strip extended past the sampled design, matching Sect. 4’s finding that the invariant direction is unconstrained at every magnitude — because the strip extent enters the ground truth $[L^*, U^*]$ and therefore every method’s coverage, not only the band’s. The narrowed-strip variant is reported in the text rather than mixed into this table (`scripts/make_table7.py` prints both).

Surface	Point calibration	Union bound	GP simultaneous band
Flat	0.92 / 0.05 / 0.53	1.00 / 0.00 / 0.24	1.00 / 0.00 / 0.14
Linear	0.24 / 0.36 / 0.71	0.99 / 0.00 / 0.29	0.95 / 0.00 / 0.15
Dip at a sampled setting	0.00 / 0.88 / 0.45	0.98 / 0.00 / 0.03	0.97 / 0.00 / 0.04
Dip between settings	0.00 / 0.85 / 0.43	0.22 / 0.17 / 0.21	0.91 / 0.00 / 0.08

the quantitative form of “more evaluation makes a wrong conclusion more confident”.

(ii) The union bound attains its guarantee, and only its guarantee.

Where the extremum of $\Delta(z)$ falls on a sampled setting it has coverage ≥ 0.98 with error rate 0 and the highest power of any valid rung (declaration rate 0.98 on the flat surface at $\Delta_0 = 0.3$, $n = 96$). Where the decisive feature falls between sampled settings it fails outright: coverage 0.22, error rate 0.17.

(iii) The GP band’s observed coverage is at least 0.91 on all four families, with error rate 0, and it pays for that in power.

We are careful with the word. These are empirical coverage frequencies over 300 repetitions per cell, not a proved guarantee, and 0.91 is below the nominal 0.95 the construction aims at. The Monte Carlo standard error at 300 repetitions is about 0.017 near a coverage of 0.9, so the 0.91 cell sits roughly two standard errors below nominal and the shortfall is unlikely to be simulation noise alone. The band’s validity is *conditional on its smoothness prior*; what Table 8 reports is how often that prior was good enough on these four surface families, which is evidence about the families we chose rather than a distribution-free statement. Its declaration rate on the flat surface at $\Delta_0 = 0.3$, $n = 96$ is 0.47, and barely grows from $n = 48$ to $n = 96$, because the band is evaluated half a grid step beyond the sampled range and the posterior diverges at the ends.

Narrowing the strip to the sampled range raises that rate to 0.77 and restores monotonicity in n , and it is tempting to adopt it on

that basis. We report it as a sensitivity analysis instead, because the change is not confined to the band. The strip extent enters the ground truth $[L^*, U^*]$, so narrowing it narrows what every method is asked to cover: point calibration’s maximum error rate on the linear surface falls from 0.36 to 0.04 and the union bound’s mean declaration rate from 0.29 to 0.19, purely from the redefinition. Comparing the band’s 0.91 observed coverage under one truth against its 0.85 under another is therefore not a like-for-like measurement of one estimator, and we do not present it as one. Within the narrowed configuration, taken whole, the band’s minimum observed coverage is **0.85** on the between-settings dip with a nonzero error rate (0.01) — further below nominal than the headline table’s figure, and we attach the ≥ 0.91 observation only to the configuration Table 8 reports.

On the real data of Fig. 6 the situation is different and the choice is forced: the 13 design points reach $|u|, |v| = 2$, and a verdict read off the band beyond that would be extrapolation with no observation behind it. Fig. 6 therefore evaluates the band over the sampled range only, and says so.

The highest-weight length scale is $\ell = 0.5$ on all three tasks, with signal scale 0.02 on carrot and 0.05 on spoon and eggplant, so the band there does not rest on a per-task tuning choice.

One scope limitation has to be stated before Fig. 6 is read alongside anything else in this paper, because it is easy to miss and it invalidates the obvious comparison. The controller plane needs 13 design points spanning stiffness and damping independently, and the only sweep we have with that design is the *pre-fix* one: an older inference-stack build, 48 episodes per condition, and per-episode rather than per-configuration standard errors.

Every other number in this paper is census-scope on the current build at $S = 5$. Fig. 6 is therefore an **exploratory analysis on a different and superseded data scope**, not a rung evaluated on the same footing as Table 6, and we do not cross-validate the two against each other. Concretely: on spoon the GP band over that older sweep declares $([+0.03, +0.43])$ while the union bound over the *same* older sweep abstains with a lower bound of exactly 0.000; on the current census (Table 6) the spoon union bound is $[+0.04, +0.56]$ and declares. Those three numbers come from two different data scopes and one cannot be read as a methodological disagreement with another. What Fig. 6 contributes is a picture of the response surface over a plane no current sweep covers — including the ratio direction that Sect. 4.4 shows the calibration data care about — and that is the only weight we place on it.

We also report a configuration we **rejected**, because it is a methodological point rather than a tuning detail. Imposing a lower bound on the GP length scale equal to the sampling spacing doubles power again (flat-surface declaration 0.22 to 0.40, linear 0.17 to 0.30) but assumes the surface is smooth at exactly the scale where the dip families are not, and coverage collapses to 0.45–0.83 with error rates up to 0.10. A response-surface method’s power can be bought only with prior assumptions, and the stronger the prior the more brittle the method is to the features it did not sample. The flatness gate we removed is the same lesson: with 13 design points the GP already fits the sampled-dip surface, so the gate only lowered coverage on the between-settings surface (0.91–0.98 to 0.89) and mis-fired on flat surfaces 16–20% of the time.

8.3 Stability on real replicate data

Synthetic ground truth cannot tell us whether the real surfaces resemble any of the four families. So we also score the criteria on real data by re-evaluation: a verdict is *stable* if a second evaluation of the same policy pair returns it. We avoid calling these “independent replicates of the same evaluation”, because the two sides of a pair differ by more than sampling noise. Some differ in the policy seed set only; others differ in the simulator port or the inference-stack build, which Sect. 5.5 shows is itself a parameter of

consequence. What the table therefore measures is consistency under changes of seed, implementation and protocol together — which is the practically relevant notion, and a weaker one than replication in the statistical sense.

The stability column rises from 0.50 to 0.88 to 1.00 down the table, and read alone it would say that conservatism buys reproducibility. It does, but not in the way that ordering suggests, because *stable* counts agreement to abstain. The all-axes row reaches 1.00 by **never declaring anything** on this data — the degenerate end of the conservatism dial, where a criterion that always abstains is perfectly reproducible and perfectly useless. The controller-axis row’s 0.88 is seven abstain–abstain agreements out of eight; it declared on exactly one pair, and that declaration did *not* reproduce.

This data therefore supports a narrower claim than the column ordering suggests. What we can say is that the union bound *abstains* more consistently. Point calibration declared on five of the eight pairs; on four of those five it declared on one replicate and abstained on the other, and the union bound turns all four into agreed abstentions, which is what drives the unsupported rate from 0.50 down to 0.12. What we cannot say, and do not claim, is that its declarations replicate better. With one declaration to its name and that one unreplicated, the affirmative rate in the last column (0 of 1, against point calibration’s 1 of 5) is not an estimate of anything; neither number has the sample size to be ranked against the other. Eight replicate pairs spanning three tasks, six of them sharing policy data, on a superseded data scope, cannot settle declaration reliability, and we do not ask them to: our claim about the validity of declarations rests on the synthetic ladder of Sect. 8.2, where ground truth exists and coverage is measured directly. What Table 9 contributes is the trade-off in the middle row — a real gain in agreement (0.50 to 0.88) bought with a real loss in declarations (0.38 to 0.06).

8.4 The delimiting experiment: a pair the real robot reverses

Everything to this point is internal to the simulator. The criterion is conservative where we can check conservatism and stabilises verdicts where we can check stability, but neither test asks whether it protects against being *wrong about the*

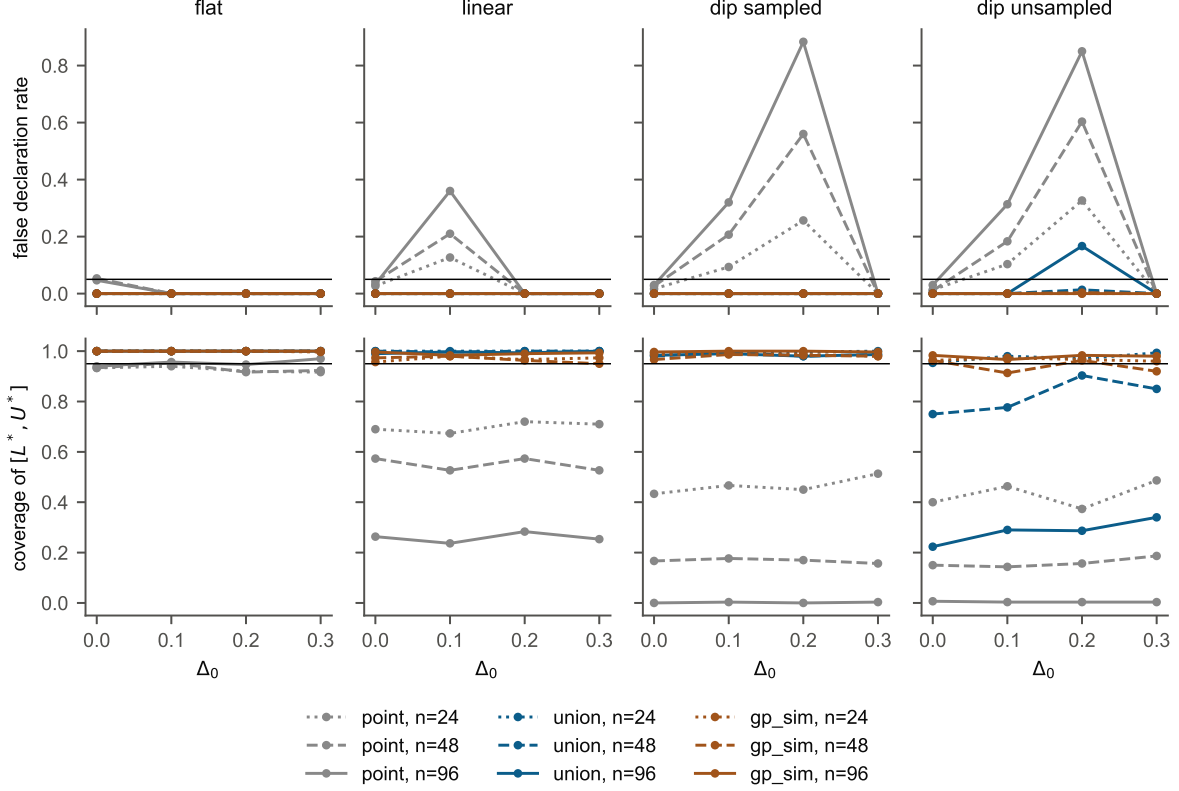


Fig. 5 The three criteria on synthetic ground truth, by response-surface family (columns), true margin Δ_0 (horizontal axis) and episode budget (line style). **Top row:** false declaration rate, with the nominal 0.05 level marked. **Bottom row:** coverage of $[L^*, U^*]$, with the nominal 0.95 level marked. Point calibration (grey) gets *worse* as the budget grows on every non-flat surface — its false declaration rate reaches 0.88 at $n = 96$ on the sampled-dip surface, and its coverage falls towards zero. The union bound (blue) holds at ≥ 0.98 coverage except on the between-settings dip, where it drops to 0.22 and declares falsely at up to 0.17; the GP simultaneous band (orange) attains an observed coverage of at least 0.91 on all four families, which is below the nominal 0.95 and is a measured frequency, not a guarantee. Drawn from the same cached cells as Table 8 and therefore on the same strip extent; the narrowed-strip variant discussed in Sect. 8.2 is not plotted, and neither is the withdrawn flatness gate.

real robot. The published tables let that be asked without hardware, on a case we selected — and we state the selection, because it is development information and not a held-out test.

The reference implementation publishes real-robot and simulated success rates for the same policies on the same task. Auditing all of them (`research-audit/published-pairwise-audit.md`), 62 within-task policy pairs have both rates published once RT-2-X is excluded for lack of public weights, and **5 of the 62 have opposite mean directions** in simulation and reality, with a further 3 tied in at least one domain. So this is not a unique case, and an earlier version of this paper wrongly called it one. We study `rt-1-converged` vs `rt-1-15pct`

on pick-coke-can (Brohan et al., 2023) for three stated reasons: both checkpoints are public, it is on the embodiment and task where we ran a complete 300-configuration census, and simulation declares it with a clear margin (+0.147). An earlier version of this paper also claimed it has the largest published real margin of the five. It has the *smallest*: the five real margins are 0.333, 0.185, 0.167, 0.148 and, for this pair, 0.067. That cuts against the case rather than for it, and the objection it raises is met below rather than omitted. These are reasons of convenience and coverage, not of significance — the five are directional disagreements between *rounded published means*, with no per-cell trial counts and no intervals on either side, so none of them is an established reversal.

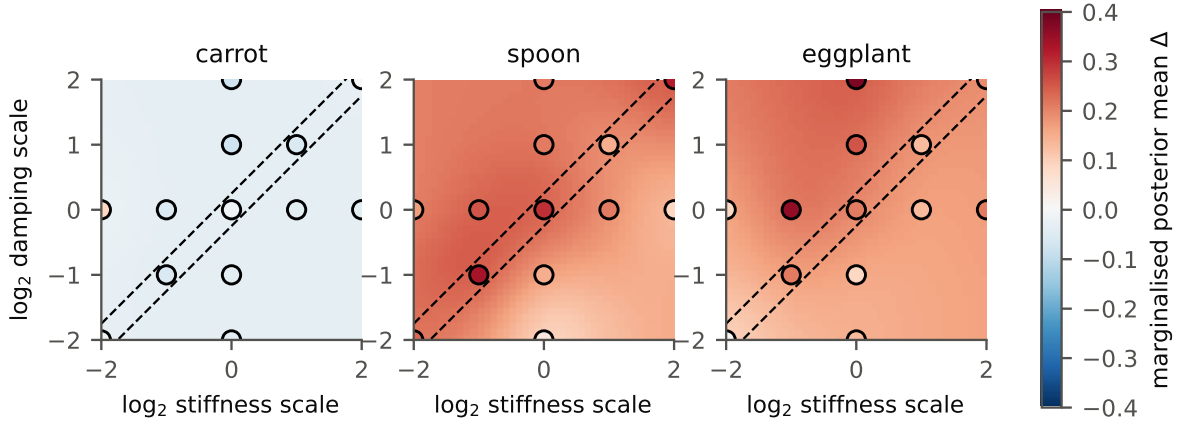


Fig. 6 Posterior mean of the policy difference over the controller plane, for three bridge tasks, marginalised over the GP hyperparameter grid; circles are the 13 sampled design points and the dashed lines bound the replay-compatible strip (gain ratio ≈ 1). The surface drawn and the bounds quoted are one computation — the restricted simultaneous band adopted in Sect. 8.2, evaluated over the sampled range $|u|, |v| \leq 2$ with 4,000 posterior function draws. Bounds and verdicts: carrot $[-0.10, +0.03]$, abstain; spoon $[+0.03, +0.43]$, octo-small better; eggplant $[-0.06, +0.36]$, abstain. The highest-weight hyperparameters are $\ell = 0.5$ with $\sigma_f = 0.02$ (carrot) and 0.05 (spoon, eggplant). **Data scope: this figure uses the older pre-fix sweep** (48 episodes per condition, per-episode standard errors), the only design that varies stiffness and damping independently. It is exploratory and is not on the same footing as Table 6’s census results; see Sect. 8.2. Generated by `analyze_response_surface.v2.py`, which prints these bounds so caption and analysis cannot diverge.

Table 9 Verdict stability over the eight replicate pairs whose two sides enumerate the same configuration grid. These eight are *not* eight independent experiments: six are policy pairs drawn from the same four Octo variants on the eggplant census and therefore share policy data, and the other two are spoon and carrot, so the eight pairs span three tasks. The underlying records are the older 48-episode pre-fix sweeps with a per-episode bootstrap, not the current census scope. *Stable* counts agreement of any kind, including agreement to abstain, so the last two columns separate out what a reader who wants to trust a declaration needs: how many pairs carried a declaration on at least one replicate, and how many of those had both replicates declare the same ordering. At most one declaration successfully replicated under any criterion (`analyze_declaration_replication.py`).

Criterion	Pairs	Stable	Contradict	Unsupported	Declare	Declaring pairs	
						n	replicated
Point calibration	8	0.50	0.00	0.50	0.38	5	1 (0.20)
Union bound, controller	8	0.88	0.00	0.12	0.06	1	0 (0.00)
Union bound, all axes	8	1.00	0.00	0.00	0.00	0	n/a

The question is whether the union bound *abstains* on a pair where the simulator’s ordering disagrees with the published real one. It is the sharpest test of the criterion available to us, because here there is an external number to disagree with — though that number is a point estimate without an interval, which we return to below. We ran both checkpoints over the complete 300-configuration census under nominal conditions plus two calibration-invisible conditions — the torque limit and object friction — for 1,800 episodes total.

One caveat about that choice of conditions belongs here rather than in a footnote. The replay evidence of Sect. 4.3 is on *WidowX* BridgeData demonstrations, and this experiment is on the Google Robot embodiment. We carried the two conditions over because they are the ones that moved verdicts on bridge, not because they are established as calibration-invisible for this arm: we have no Google Robot replay set, so for this embodiment “calibration-invisible” is an assumption imported from the other, and the friction axis is in any case unobserved by free-space replay on either. The experiment should be read as a test of

Table 10 The real-versus-simulation reversal pair. Our reproduction of the published simulated rates is $+0.006$ and $+0.007$, so the disagreement is reproduced faithfully rather than introduced.

Source	rt-1-converged	rt-1-15pct	Margin
Published real robot	0.853	0.920	-0.067 (15pct better)
Published simulation	0.857	0.710	$+0.147$ (converged better)
Our nominal census	0.863	0.717	$+0.147$

the criterion’s behaviour on a pair with external ground truth, not as a propagation of a verified compatible set.

The result is adverse. Point calibration on the nominal condition gives $[+0.083, +0.210]$, declaring **rt-1-converged** better. The union bound over the two dynamics conditions gives $L = +0.070$, $U = +0.240$ — **also declaring rt-1-converged better**, which is the ordering the published real-robot rates reverse.

Widening from the point to the set moved the lower bound from $+0.083$ to $+0.070$, so the shift that would reach abstention is 0.070 : the bound has to lose that much on its lower end, not the 0.214 separating the simulated margin from the real one. Measured against 0.070 , the two dynamics conditions are not close — their margins span 0.133 to 0.177 , a spread of 0.044 , which is less than the distance to zero. But that is a statement about *these two conditions*, not about hidden variables in general, and the distinction matters: across the twelve texture $\times$ condition cells the margins span $+0.053$ to $+0.213$, a spread of 0.160 , more than twice what abstention requires. The axis that carries no physics has the range to move this verdict; the two dynamics parameters do not.

Having the range to move the verdict is not the same as rescuing the criterion, however, and on inspection it does not. Because this run puts a second policy on the same 300-configuration grid, it also settles the question left open by Sect. 5.5: whether a visual variant that carries no physics can flip a *ranking* rather than merely inflate a single policy’s variance. Adding **urdf_version** as a fourth set dimension (12 cells of 75 configurations each) does turn the verdict to abstention, $L = -0.067$ — but the abstention is of the second kind identified in Sect. 7.5. All twelve cells have **positive** point margins, from $+0.053$ to $+0.213$, and not one places **rt-1-15pct** above **rt-1-converged** with interval support. The

bound reaches zero only through interval width at 75 configurations per cell.

We put this more carefully than we first did, because the asymptotic statement is easy to overclaim. Each cell’s interval is a policy-noise interval over an enumerated 75-configuration sub-grid, so its width shrinks with additional *runs* per configuration rather than with additional episodes. If the twelve observed point margins were the population quantities, L would tend to their minimum, $+0.053$, and the abstention would dissolve. But they are not: with one run per configuration each cell margin carries binomial uncertainty we have not resolved, so twelve positive *sample* means do not establish that all twelve population means are positive, and $+0.053$ is where the bound tends under our estimand rather than a measured endpoint. What we can say without that assumption is narrower and enough for the purpose: the abstention here is produced by interval width and not by any cell disagreeing in sign, so by our own test in Sect. 7.2 it is at least partly our power rather than a demonstrated parameter fact, and we decline to claim it as a success of the criterion. Resolving whether it is *only* power would need repeat runs per configuration in each cell, which we have not done.

One objection should be met here rather than deferred. The published real margin is -0.067 , which is of the same order as the policy-seed noise we measure in simulation, and the published real rates carry no interval of their own — so the reversal might be real-robot measurement noise rather than a model gap. The reading is asymmetric, and either branch supports the same conclusion. If the reversal is a genuine model gap, the criterion declared an ordering that is wrong. If it is real-robot noise, then the real evaluation cannot resolve this pair either, and a simulated declaration still carries no warrant about the robot. What the criterion may not do, on either branch, is license the sign it produced.

The conclusion we draw is therefore about the criterion’s reach, and we bound it to what was actually tested. The difference between the simulated margin and the published real one is 0.214, and one policy’s own rate differs by 0.203 (`rt-1-15pct`: 0.920 real against 0.717 simulated). Against that, abstention on this pair would have required the union’s lower bound to lose 0.070. The three dynamics cells we ran span 0.044 and do not reach it; the twelve texture $\times$ condition cells span 0.160 and do reach it, though only by interval width and without any cell changing sign. So the honest statement has three parts. *Within the settings we ran*, no configuration of the calibration-invisible parameters produces the real robot’s ordering. *Beyond them* we do not know: we tested two dynamics conditions at one value each and one visual axis at its four values, out of a parameter family that is continuous in friction, density, the common gain scale and the torque limit, and we did not explore interactions. And the comparison target is a point estimate — the published real rates come without intervals, so the 0.214 is a difference between one number with an interval and one without.

We therefore claim only this: on the conditions tested, the criterion declared the ordering that the published real-robot point estimates reverse. Whether the cause lies outside the whole parameter family that demonstration replay can constrain is a stronger statement that would need either wider coverage of that family or a real-robot replication with its own uncertainty, and we do not make it. In the ledger of Sect. 3.4 the finding is that b_{ij} is plausibly large on this pair and that a criterion built on E_θ and E_{MC} cannot see it — which is enough to delimit the criterion, and is all we need it to do.

8.5 What the ladder is for

Taken together the three tests place the criterion precisely, and the placement is narrower than we set out to establish. It bounds the ambiguity the calibration data leaves in the parameters it can represent: on synthetic ground truth its observed coverage is conservative on all four surface families where its smoothness prior holds and fails where it does not (Sect. 8.2); on real replicates it *abstains* consistently where point calibration declares on one replicate and not the other, which

is a claim about abstention and not about declarations (Sect. 8.3); and it changes the verdict on 3 of 17 bridge policy pairs at the reference protocol’s three seed sets, falling to 1 of 17 at five seed sets over a full census (Sect. 7.2). It does not bound, and cannot bound, the ambiguity arising from a model gap that no setting of those parameters expresses — and Sect. 8.4 exhibits a pair where reaching abstention would have needed 0.070 while the gap between the simulated and published real margins is 0.214 and points the other way.

We therefore do not recommend the union bound as a device for deciding which policy is better. We recommend it as a device for detecting when a benchmark’s own evidence cannot decide, which is a weaker and more defensible role, and for which Sects. 8.2 and 8.3 are the relevant evidence. A declaration from it is not a sim-to-real guarantee; Sect. 8.4 is the counterexample that makes that sentence necessary.

9 Limitations

We list these in descending order of how much they should change a reader’s confidence, rather than in the order that flatters the paper.

1. **The operating-point measurement covers one task and one policy pair.** Section 4.4 finds that the replay loss is minimised at ($k \times 2, d \times 0.5$, delay 1) on both stacks, and that nominal — ratio 1, no delay, where every published rate is computed — is rejected against that minimiser. Section 7.7 measures what that costs on the eggplant census and finds the point verdict flipping from abstention to a declaration while the set verdict abstains at both. That is one task, one policy pair, and one alternative operating point. We do not know whether the pattern — point verdicts sensitive to the operating point, set verdicts not — generalises, and a single instance cannot establish it; it is the cheapest available test of the question and it is not a survey. Extending it means the same census at the fitted setting on spoon and carrot, and on more policy pairs.
2. **The fibre through the fitted point is not yet censused to full width.** Fibre *membership* at the fitted ratio is measured rather than assumed: the replay grid varies the common scale at seven other ratios but only once at

0.25, so we swept the scale sixteenfold at ratio 0.25 directly and re-measured the torque limit at the fitted gains (Sect. 7.7). What remains open is policy data, not replay data: four of the six planned conditions in the fitted fibre have a complete 64-configuration census and two are still running. The envelope is a min/max over conditions and so only widens, and it already abstains, so the two outstanding conditions cannot change the set verdict — but they can widen it, and the number we report will move.

3. The criterion’s central negative result rests on one selected policy pair.

Section 8.4 shows the union bound declaring an ordering the published real means reverse, on one pair, one task, one embodiment. It is *not* the only such pair: our audit of the published tables finds 5 of 62 within-task pairs with opposite mean directions, and we chose this one for public checkpoints, census coverage and a clear simulated margin — not for the size of its real margin, which is the smallest of the five. That makes it a case selected with development information in hand, which is the right way to read it — a demonstration that the criterion’s reach is limited, not an estimate of how often the limit is reached, and not a held-out test. The other four are available to anyone wishing to look, and one of them (*octo-base* vs *octo-small* on carrot) is a bridge pair we already have a census for. A reader should conclude that a union-bound declaration carries no sim-to-real warranty, and should not conclude that such declarations are usually wrong.

4. **No hardware of our own.** Every simulated number here is ours; every real-robot number is quoted from the reference implementation’s published tables. Those are the values the benchmark itself is validated against, and using them is what makes Sect. 8.4 possible without a laboratory, but they carry their own uncertainty and are published without intervals: they are single-protocol real evaluations, and the pair in Sect. 8.4 has a real margin of 0.067, of the same order as the seed noise we measure in simulation.

The path to a real-robot test is short enough to specify rather than gesture at, and we specify it here as a *plan*, not as work done. The experiment that would falsify or confirm the one declaration this paper’s criterion gets wrong is:

rt-1-converged versus *rt-1-15pct* on pick-coke-can, on a Google Robot or equivalent 7-DoF arm, at the published protocol’s scene layout, with $n \geq 200$ trials per policy so that a real margin of 0.067 is resolved rather than assumed. At the published rates 0.853 and 0.920 a normal-approximation interval on the difference has half-width 0.062 at $n = 200$, 0.044 at $n = 400$ and 0.031 at $n = 800$, so $n = 200$ is where the published sign becomes distinguishable from zero at all and $n \approx 400$ where it is resolved with margin. Two things must be frozen before the trials start, because both are choices we have shown decide answers: the analysis rule (interval, threshold and observation unit) and the simulator operating point against which the comparison is made. Our release contains the simulated side at census scope, so the real side is the only missing term. The four other reversal pairs in the 5-of-62 audit of Sect. 8.4 extend the same design at the same cost per pair, and one of them — *octo-base* versus *octo-small* on carrot — is a bridge pair for which we already hold a complete configuration census, so it needs only the real arm.

We are stating a protocol we cannot execute: we have no arm, and the two public evaluation services that would have supplied one were unavailable during this work (Sect. 1.3). A reader with either should be able to run the above from the release without asking us anything.

5. One suite, and two kinds of conclusion.

Every measurement here comes from a single evaluation suite, and the two kinds of claim we draw from it do not travel equally. Claims that follow from *construction* should hold for any benchmark built the same way: that an index-driven initial state makes the configuration set a finite population rather than a sample; that free-space demonstration replay is uninformative about a common gain scaling and about an inactive saturation limit; that a criterion over a compatible set bounds only the ambiguity that set can express. Those arguments nowhere depend on which suite, policies or tasks we used. The *numbers* are specific to this instance and should not be carried anywhere — 0.121 for the torque shift, 0.400 for the texture spread, 1 of 17 disagreeing pairs, the 0.038 reproduction.

A reader should take the mechanisms as general and the magnitudes as an existence proof that they can be large.

6. **Narrow coverage of policies, embodiments and tasks.** Three model families (two Octo variants with three deployment configurations, OpenVLA, two RT-1 checkpoints), two embodiments and four tasks. OpenVLA is run 4-bit quantised, which we did not control against a bf16 baseline; the torque mechanism in Sect. 7.4 is a claim about the quantised model as deployed. The dependence of decision-relevance on the task-policy pair means our coverage is exactly the wrong shape for extrapolation: we show the effect exists on one task and is nearly absent on another, and we cannot say which is typical.
7. **The strong form of the non-identifiability claim is false in our data.** No calibration-invisible parameter reverses a sign-determined ranking anywhere we looked, and the one between-condition sign reversal we once claimed did not survive a larger budget (Sect. 7.5).
8. **The fibre is sampled, not covered.** The union bound evaluates a discrete set of parameter settings, so its guarantee is of sampling-extremum type, and Sect. 8.2 shows it failing (observed coverage 0.22) precisely when the decisive feature lies between sampled settings. Concretely, we sample the common scale at $\times \frac{1}{4}$ and $\times 4$, the torque limit at $\times \frac{1}{2}$, and friction and density at one value each — five points standing in for a region that is continuous in four coordinates, with no interactions explored. On the contact axis three sampled points are too few for the GP rung to be trustworthy, so those axes use the union bound and inherit its limitation.
9. **Low success rates produce many abstentions, and some are ours rather than the data’s.** Several tasks sit at success rates of 0.1 to 0.5, where even a complete 24-configuration census at three seeds gives half-widths of 0.05 to 0.13. We applied the $h \rightarrow 0$ test wherever the budget allowed, but for the original-stack rows of Table 6, which have one run per configuration, we cannot.
10. **Implementation-build drift is separated on two policies only.** The -0.109 shift and the 0.73 – 0.88 episode-level agreement come

from a single controlled re-run of two Octo policies across two inference-stack versions — one run of each build, so the figure carries no seed replication and no interval. We treat π as an axis of the same kind as z on that basis, which is thinner evidence than the rest of Sect. 5 rests on.

11. **Part of the platform work is exploratory.** The ManiSkill3 configuration was used for the sweeps, with the original reference stack used for cross-checking; the two agree on the replay invariances to sub-millimetre and on the union-bound verdicts in Table 6, but they differ on point verdicts for one task, and their eggplant configuration grids differ so that per-episode comparison is undefined there. Rows are labelled by stack throughout and should not be pooled.

10 Conclusion

We asked which ranking claims a calibrated simulation benchmark’s own evidence determines, and answered it on a widely used sim-to-real evaluation suite. The calibration protocol is uninformative about a common scaling of the joint gains, and all but exactly so about the torque limit — at most 3.1×10^{-6} of composite loss across a sixteenfold rescaling, and bitwise identical replays on 97 of 98 demonstrations, on two independent simulator stacks. It is *not* uninformative about the controller ratio and the execution delay, and there it rejects the setting the simulator ships with: the benchmark is operated at a point its own calibration evidence excludes, and moving to the point that evidence prefers shifts a policy difference by 0.031, which is enough to turn a point-calibration abstention into a declaration while the set-valued verdict abstains at both. The benchmark’s initial states form a finite population of 24 to 300 configurations rather than a sample, so enumerating them sets configuration sampling error to zero, and two widely used ports enumerate different populations. And following the published protocol, down to the pseudo-random-number lifecycle its wrapper implies, reproduces its rates to 0.038 mean absolute difference, one-signed across the eight cells, while showing that its own 72-episode budget resolves one of the four orderings it reports

— two under a coarser variance model, a sensitivity we report rather than choose between on the strength of the conclusion.

The practical message these findings converge on is about what evaluation budget can and cannot touch, and we state it as that rather than as a ranking of magnitudes. On the eggplant census a single seed set gives a half-width of ± 0.132 on Δ , the reference protocol’s three seeds ± 0.076 , and ten seeds would give ± 0.042 : this term falls as $1/\sqrt{S}$ and a practitioner can buy it down. The others do not move with budget at all. A halved torque limit, invisible to calibration, shifts Δ by 0.121; the robot texture variant spans Δ by 0.160 across four settings that carry no physics; the implementation build shifts it by 0.109. These are shifts and spans rather than dispersions, so we do not say which is “biggest” — the point is that **no number of extra episodes reduces any of them**. A practitioner who responds to an ambiguous benchmark result by running more episodes is buying down the one term that responds to episodes, and — on any non-flat response surface — is making a possibly wrong conclusion more confident while doing so.

What we recommend follows from where each term sits. Report the estimand and the budget, because a rate difference means something different as a benchmark value over an enumerated population than as a generalisation value over sampled scenes, and the correct interval differs accordingly. Enumerate the configuration grid, since it is finite and the marginal episode past it carries no information. Report which implementation build and which visual variant a number came from: the aggregation over variants is part of the documented protocol, but a number reported without naming the variant cannot be told apart from a mean over all four, and on our census those differ by up to 0.227. Report the operating point *and how it stands against the calibration objective*, which costs nothing and which no suite we know of does. A benchmark evaluated at a setting its own replay data reject is not thereby wrong, but a reader cannot tell whether the verdict computed there is the verdict at the setting that data prefer — and on the one pair where we checked, it is not. The direction is worth stating precisely: nominal abstains and the fitted point declares, so what moved was not a declaration failing to survive but an abstention becoming one, on

a change in $\hat{\Delta}$ that is itself unresolved (Sect. 7.7). No criterion defined over the calibration-invisible directions can supply this, because the discrepancy lies in the directions the calibration *does* constrain. Spend budget on seeds before episodes — but only for policies that consume a seed. For a deterministically decoding policy the seed axis is degenerate (Sect. 5.3), a census is already exact, and additional runs buy nothing at all; there, the only way to reduce uncertainty is a larger configuration grid. And use a set-valued verdict — but for the role it can actually fill.

That last point is where our own expectations were corrected. We built the compatible-set criterion expecting to show that it repairs unreliable rankings, and what we can show is narrower: it detects when a benchmark’s evidence cannot decide, it *abstains* consistently where point calibration declares on one replicate and not the other, and it changes the answer on 3 of 17 pairs at the reference protocol’s three seed sets — falling to 1 of 17 when we spend five seed sets over a full census, which is a larger evaluation budget than any published one we are aware of and not a tightening of the data-retention rule. We are careful about the middle claim: on our eight replicate pairs the criterion declared once and that declaration did not reproduce, so what the replication data support is reliable abstention, and the validity of its declarations rests on the synthetic ground truth of Sect. 8.2 rather than on replication. It does not protect against a model gap that no value of the parameters expresses, and on a selected pair — one of five in the published tables whose simulated and real means point opposite ways — it declares the ordering those means reverse. Because the published real rates carry no intervals, we cannot say the criterion is wrong about the robot there; what we can say is that its declaration carries no warranty about the robot, which is the weaker claim and the one the evidence supports. A criterion of this kind bounds the ambiguity it can represent, and on at least one pair that is not the ambiguity that decides the answer.

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Use of generative artificial intelligence.

A large language model (Anthropic Claude) was used to assist with writing and revising the manuscript, and to help write the analysis and figure-generation scripts. The study design, the experiments and the interpretation of results are the authors' own; all quantitative results are computed from the episode records by the released scripts. The authors have verified the content and take full responsibility for it, and no language model is listed as an author, in accordance with journal policy.

Declarations

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Ethics approval and consent to participate. Not applicable. This study involves no human participants and no animals.

Consent for publication. Not applicable.

Data availability. All episode-level records underlying the results are openly available in the archived release at <https://doi.org/10.5281/zenodo.22893458>, which resolves to the newest version and lists every version with its own DOI. **The results in this paper correspond to version v1.5.0 and to that version's DOI,** not to the concept record generally: v1.0.0–v1.1.3 shared one immutable set of records, but v1.2.0 onwards added evaluation data — the reference-lifecycle reproduction, the fitted-operating-point censuses and the additional replay sweeps — so the versions are no longer interchangeable and an earlier one will not reproduce the tables here. Development is at <https://github.com/Jun-Jason-Ji/benchmark-ranking-determinacy>. The records are append-only JSON Lines files, one per policy, task and simulator condition, together with the per-run provenance log that records the port branch, inference-stack version and random seeds for every episode, and a `SHA256SUMS.txt` whose header states the file count for that version. The real-robot reference values used in Sect. 8.4 are the

published values distributed with the benchmark's own source and are not ours to redistribute; the manuscript cites their location.

Materials availability. Not applicable.

Code availability. The evaluation harness, the configuration-census utilities, the six resumable evaluation queues, the analysis scripts that generate every table and figure in this paper, and the Vulkan compatibility layer described in Appendix B are openly available in the same archived release, <https://doi.org/10.5281/zenodo.22893458>, under the MIT licence for original code; evaluation records and outputs derived from third-party simulators, policies and demonstration data are governed by the NOTICE file in that release. `REPRODUCIBILITY.md` gives the commands that regenerate every table and figure from the episode records.

Author contributions. **Jun Ji:** Data curation, Methodology, Formal analysis, Resources, Writing – original draft. **Bowen Tan:** Software, Formal analysis, Writing – review & editing. **Yi Li:** Supervision, Project administration, Writing – review & editing. **Xiaolei Zhang:** Supervision, Project administration, Writing – review & editing. **Shengjie Guo:** Supervision, Project administration, Writing – review & editing. **Yi Sui:** Supervision, Validation, Writing – review & editing.

Appendix A Test inversion for the compatible set

A.1 Construction

Calibration observes M real demonstrations. For a candidate parameter z , let $L_m(z)$ be the replay error on demonstration m — the deviation between the simulated and recorded trajectories — and let $\hat{z}$ minimise $\bar{L}(z) = M^{-1} \sum_m L_m(z)$ over the grid we search.

Rather than report $\hat{z}$, we invert a one-sided test. For each candidate z we form the paired per-demonstration differences $\{L_m(z) - L_m(\hat{z})\}_{m=1}^M$, resample demonstrations with replacement B times, and take $\text{lo}_{1-\alpha}(z)$ to be the α quantile of

the bootstrap distribution of their mean. Then

$$C_\alpha = \{z : \log_{1-\alpha}(z) \leq 0\}, \quad (\text{A1})$$

the set of parameters whose replay error is not significantly worse than the best. Pairing by demonstration matters: the demonstrations differ enormously in difficulty, and an unpaired test compares that between-demonstration spread against the parameter effect rather than against itself.

The threshold in Eq. (A1) is exactly zero, with no equivalence margin, so *any* strictly positive lower bound rejects however small it is. That is a consequential choice and we apply it without exception, including where it costs us: Sect. 4.4 reports that it rejects the simulator’s own nominal setting on both stacks. `rebuild_compatible_set.py` exposes the search domain, the loss, α and an optional margin $t \geq 0$ as arguments, and prints the margin that would be needed to retain any given candidate, so that if a future version of this analysis wants an engineering-equivalence tolerance it has to name one and defend it rather than inherit it from a reference choice.

Two properties of this construction carry the argument of Sect. 4, and we state them with the measured numbers rather than in the limit.

The insensitive directions move the loss by six to ten orders of magnitude less than the identified ones.

One clarification before the numbers, because two different tests appear in this paper and mixing them is easy. The set construction of Eq. (A1) tests candidates against the grid’s *loss minimiser*, and under that test nominal is rejected (Sect. 4.4). The figures below are a different and narrower comparison: each insensitive direction against *nominal*, which measures the flatness of the loss along that direction and says nothing about whether nominal itself is retained by the set construction. It is not. What follows is therefore evidence about the directions, not about the point they pass through.

If $L_m(z) = L_m(z')$ for every m , the paired differences vanish identically, the bootstrap distribution is a point mass at zero, and $\log_{1-\alpha} = 0 \leq 0$ irrespective of α , B or M . Our two directions approach that limit rather than attaining it,

so what matters is the measured margin. Tested against nominal on the 98 demonstrations:

- *Common gain scaling.* Over the fourfold range $\times 0.5$ to $\times 2$ at fixed ratio and delay, the paired mean loss differences are at most 1.6×10^{-7} with 95% lower bounds no lower than -5×10^{-7} . Retained.
- *Torque limit halved.* The paired mean loss difference is -1.1×10^{-10} on ManiSkill3 and -1.8×10^{-8} on the original stack, with exactly one of 98 demonstrations contributing a nonzero term on each. Retained.

For comparison, the identified directions are rejected with lower bounds of $+1.5 \times 10^{-3}$ to $+2.2 \times 10^{-3}$ (Sect. 4.4). Every loss figure in this appendix is in the composite units of Eq. (3), which adds metres to radians and is therefore not a length. The separation between “retained” and “rejected” here is four to seven orders of magnitude, which is why the conclusion does not turn on the choice of α , of B , or on whether one demonstration in 98 behaves differently. What it does turn on is the reference point, and that is the lesson of Sect. 4.4: tested against the grid minimiser rather than against nominal, the same procedure retains the same invariant *directions* while rejecting the *point* they pass through.

Unobserved parameters are compatible for a different reason.

Contact friction and object density never enter a free-space replay trajectory at all, so their compatibility reflects the protocol’s *coverage* rather than its precision. We flag these separately throughout, because a reader might otherwise conclude that a longer or better-excited demonstration set would constrain them. It would not; a different experiment is needed.

Identified directions are excluded with margin.

Along the gain ratio and the execution delay the same procedure returns strictly positive lower bounds. Relative to nominal, ratio $\times 0.5$ and $\times 2$ raise the mean paired loss by $+0.0064$ and $+0.0089$ (Sect. 4.3); relative to the grid minimiser, nominal itself is rejected at $+0.0022$ on ManiSkill3 and $+0.0015$ on the original stack (Sect. 4.4).

The construction therefore discriminates, and discriminates sharply enough to reject the setting the benchmark ships with; it is not vacuously inclusive.

A.2 The error ledger, and what this paper does not establish

Writing $[L_{ij}, U_{ij}]$ for the union-bound interval of Sect. 3.3, the claim that policy i is genuinely better in practice requires three events at once:

$$E_\theta = \{z^* \in S\}, \quad (\text{A2})$$

$$E_{MC} = \{\forall z \in S, \forall \pi \in \Pi, \forall (i, j) : \Delta_{ij}^- \leq \Delta_{ij}^{\text{sim}}(z; \pi) \leq \Delta_{ij}^+\}, \quad (\text{A3})$$

$$E_b = \{\forall (i, j) : |\Delta_{ij}^{\text{real}} - \Delta_{ij}^{\text{sim}}(z^*)| \leq b_{ij}\}, \quad (\text{A4})$$

On $E_\theta \cap E_{MC} \cap E_b$ every real difference lies in $[L_{ij} - b_{ij}, U_{ij} + b_{ij}]$ simultaneously, and *if* those events hold at levels $1 - \alpha$, $1 - \gamma$ and $1 - \beta$, a union bound gives at least $1 - \alpha - \beta - \gamma$ without any independence assumption. We write the levels conditionally because none of the three is established here — see the caveats below, of which the first is that our S carries no coverage guarantee at all. Note that E_{MC} is stated over $\Pi \times S$: the implementation build is treated as an axis of the same kind as the physical parameters, for the reason given in Sect. 5.5.

Each event needs its own argument, and five caveats are worth stating explicitly because each is a way the ledger can be misused. The first two describe gaps between what this ledger asks for and what we report; we do not close either.

1. **Neither set we use carries a coverage probability, and for two separate reasons.** The first is the selection effect of Sect. 3.2: C_α as computed picks its reference from the same demonstrations it then tests on, and under a null of equal-risk candidates at our K and M that retains a true minimiser about 53% of the time rather than 95%. So even C_α is an empirical rule, not a confidence set; a split-sample variant recovers most of the level and we report it, but we do not claim exactness for it either. The second is that our verdicts do not range over C_α at all but over the invisible fibre through nominal (Sect. 4.4), which is generated

by a point the calibration data reject. On both counts $P(E_\theta)$ is unknown for the sets we use, and every ledger statement in this paper is conditional — on the benchmark’s operating point, and on levels we do not establish.

2. **Our intervals are per pair; E_{MC} and E_b are joint events.** We compute an interval for each policy pair separately and report the collection. That is not $\forall(i, j)$: a family of per-pair statements must be inflated for multiplicity before it can enter the ledger, and we have not performed that inflation. The knife-edge bounds reported in Sect. 7.2 (+0.0091, +0.0095) illustrate the point rather than correct for it — a simultaneous procedure over 17 pairs would not retain declarations resting on margins that small. Our per-pair results should be read as exploratory in the multiplicity sense. This is a statement about *multiple policy pairs*, not about the envelope over conditions: for a fixed finite set of conditions, taking the minimum lower bound and the maximum upper bound is itself conservative and needs no further correction across those conditions.
3. **E_θ presumes that z^* exists.** The event is empty of content if no parameter setting reproduces reality — that is, if the simulator is structurally misspecified rather than merely uncalibrated. The fitted parameters may then be no more than locally effective quantities, and α bounds nothing. Section 8.4 is best read as a case where this premise is under strain: no setting of the parameters we can vary reproduces the observed real ordering.
4. **A Gaussian-process posterior standard deviation is not γ .** Substituting the posterior spread of the response-surface rung into γ without validating its calibration would assume the very smoothness the rung is there to test. Section 8.2 measures that rung’s coverage on synthetic surfaces instead, including surfaces that violate its prior.
5. **Surrogate error must be bounded to be absorbed.** Where an approximation has a certified error bound, the band can be widened by it. Where it does not, that rung is an empirical diagnostic and must not be reported as a coverage guarantee.

Two further points concern variance reduction rather than coverage. Pairing on common initial states and common exogenous random draws reduces the variance of $\hat{\Delta}$ and we use it throughout, but it does not by itself produce a simultaneous band over F_{nom} — simultaneity has to be constructed, not inherited from pairing. And the internal action sampling of two different policies does not correspond; forcing it into a pairing and treating the result as a shared exogenous environment is not valid, which is why Sect. 5.3 treats the seed as a replication axis only for policies that consume it.

What this paper establishes. Sections 5–7 measure E_{MC} , and enumerating $\mathcal{C}$ removes one of its two components exactly. Section 4 characterises E_θ , whose width dominates the ledger. We do **not** establish E_b : we have no per-pair sim-to-real bias bound, and Sect. 8.4 exhibits a pair on which b_{ij} is large enough to swamp every other term.

Appendix B Headless rendering of the original stack without a GPU

The published-protocol reproduction of Sect. 6 and the fractal experiment of Sect. 8.4 both require the *original* reference stack rather than the ManiSkill3 port, because the two enumerate different configuration grids (Sect. 5.1). That stack renders through SAPIEN’s `svulkan2`, which requests `VK_KHR_external_semaphore_fd` at device creation. Under WSL2 on Mesa lavapipe no driver offers file-descriptor-based external semaphores, so device creation fails and the renderer never initialises — on a host with no GPU there is no path that satisfies the request.

The extension is requested but, for our workload, unused: fd-based external semaphores serve cross-API synchronisation, which `svulkan2` exercises only when CUDA interop is enabled. We therefore supply an explicit Vulkan layer, `VK_LAYER_the_world_fakesemfd`, that makes the request satisfiable without providing the underlying capability. In 210 lines of C it:

1. advertises `VK_KHR_external_semaphore_fd` and `VK_KHR_external_fence_fd` in

2. strips those two names from `ppEnabledExtensionNames` at `vkCreateDevice`, so the driver is never asked to honour them;
3. removes `VkExportSemaphoreCreateInfo` and `VkExportFenceCreateInfo` from create-info chains; and
4. stubs the fd import and export entry points, which are consequently never reached.

It is an *explicit* layer, enabled per process through `VK_LAYER_PATH` and `VK_INSTANCE_LAYERS` and installed nowhere:

```
export VK_LAYER_PATH=<dir>
export VK_INSTANCE_LAYERS=\
    VK_LAYER_the_world_fakesemfd
```

so no other process on the host sees a modified Vulkan environment. It builds with a single `gcc -O2 -shared -fPIC` invocation against the Vulkan headers.

Why this does not invalidate the results.

The layer makes a capability appear present that is absent, which is exactly the kind of intervention that should be distrusted in a paper about whether evaluation results are determined by their inputs. Two things bound the risk. First, the mechanism: every path that would consume an fd-based semaphore is removed rather than faked, so the renderer takes the same code path it would take on a driver that had never advertised the extension — the layer does not substitute an approximate implementation for a real one. Second, and more convincing, the check in Sect. 6.2: this software-rendered, layer-mediated configuration reproduces the published success rates to a mean absolute difference of 0.038 across eight cells, and to +0.003 and −0.040 on the second embodiment. A renderer that differed materially from the reference would not agree to within one seed’s worth of noise. The one-signed part of that difference (−0.038) is consistent with a rendering or physics-backend gap and we do not attribute it to the layer specifically; what the check bounds is the total, which is small.

When it would be unsound.

If CUDA interop were enabled, the stubbed entry points would be reached and the synchronisation they are meant to provide would be absent. Any use of this layer with GPU interop active should be assumed incorrect until re-checked. Our runs use CPU physics with software rendering, so the path is never taken.

Appendix C Complete condition table with per-cell counts

Table C1 gives, for the pair reported in Fig. 4 and the corresponding rows of Table 6, the configurations the task defines, the configurations actually covered, and the number of observations per configuration. The intervals in the body cannot be read without these: an interval over a complete census with five runs per configuration and one over a complete census with two are different objects, and the run count is what distinguishes them. Every cell here is a complete census, so the configuration sampling error is zero throughout and the intervals carry policy noise only (Sect. 5.4).

The run counts are *not* uniform within a cell, which Remark 1’s shorthand “ $S = 5$ ” elides. Where we write $S = 5$ we mean five seed-set directories; the per-configuration episode count is five on most configurations and four or six on the rest, for the reasons the caption gives. Any statement of the form “five runs per configuration” should be read as the directory count and checked against this table.

Eq. (7) sums over the per-configuration run count, which is the “Runs/config” column rather than the “Dirs” column, and Sect. 7.2 reports the alternative directory unit and the two rows on which the two differ. The columns here are generated by the same code path as the tables, so the counts and the estimator cannot disagree.

Appendix D Reproduction steps

The release archived at the DOI in the Data Availability statement carries a REPRODUCIBILITY.md

that is the authoritative and executable version of this appendix; what follows is its structure.

Regenerating the tables and figures requires no evaluation runs. Every number in this paper is computed from the episode records rather than transcribed, so the analysis scripts rebuild the per-pair verdict table (`make_core_table.py`, with `--unit` selecting the observation unit of Sect. 7.2), the compatible set on the 98-demonstration grids against the loss minimiser (`analyze_compatible_set_v2.py`, and `rebuild_compatible_set.py --split` for the split-sample variant; the older `analyze_compatible_set.py` is superseded and its defects are recorded in its own doc-string), the operating-point comparison of Sect. 7.7 (`analyze_fitted_point.py`, whose `--selftest` exercises the completeness gate), the selection-bias check behind Sect. 3.2 (`check_selection_bias.py`), the reversal pair of Sect. 8.4 (`analyze_fractal_reversal.py`), the per-pair torque shift of Sect. 7.3 (`analyze_torque_shift_s5.py`), the build-drift figures of Sect. 5.5 (`analyze_platform_drift_paired.py`), this appendix’s table (`make_appendix_c.py`), and all six figures, which a companion script re-renders at the journal’s geometry and audits against its width and lettering requirements.

Re-running the evaluations is needed only to reproduce the records themselves. Two simulator stacks are involved and they are not interchangeable: their configuration grids differ, so per-episode comparison between them is undefined for the eggplant task (Sect. 5.1). The original reference stack is used for Sects. 6 and 8.4 and runs headless without a GPU through the layer of Appendix B; the ManiSkill3 port is used for the controller sweeps. Requirements are pinned, and the evaluation queues are resumable: each skips episode ids already present in its target record file, so an interruption costs only the episode in flight.

Four traps are worth naming for anyone recomputing. Enumerate the configuration grid rather than sampling it, since episodes past the grid size are exact repeats. Treat the policy seed as a replication axis only for policies that consume one. Pair any cross-directory comparison on shared episode ids: the record directories hold different episode counts, and because the episode

Table C1 Per-cell configuration and run counts for octo-small versus octo-base on the three bridge tasks, under the nominal condition and the five calibration-invisible conditions. “Configs” is the size of the task’s configuration grid and “Covered” how many were evaluated. “Dirs” is the number of seed-set directories holding records for the cell; “Min runs” is the smallest per-configuration run count in it; and the last two columns give the full distribution the variance estimator uses, written as runs $\times$ configurations. Those three are different numbers, and the reason they differ is worth giving because it is easy to misread the middle one as the first: seed set B holds 96 episodes on a 64-configuration grid, so ids 64–95 wrap onto configurations 0–31 and give those an extra observation, while B stops at configuration 47 for three conditions, leaving those configurations one run short. On the eggplant census the result is 48 configurations with five runs and 16 with four for three conditions, and 32 with six and 32 with five for two others — never a uniform five. Eq. (7) sums over the per-configuration S_c , so the intervals were computed from the true counts throughout; what was wrong was this table’s summary of them. Generated from the episode records by `scripts/make_appendix.c.py`.

Task	Condition	Configs	Covered	Dirs	Min runs	Runs/config, S	Runs/config, B	Δ
carrot	nominal	24	24	2/2	2/2	2 $\times$ 24	2 $\times$ 24	−0.042
	iso_x0.25	24	24	2/2	2/2	2 $\times$ 24	2 $\times$ 24	−0.083
	iso_x4.0	24	24	2/2	2/2	2 $\times$ 24	2 $\times$ 24	−0.042
	force_x0.5	24	24	2/2	2/2	2 $\times$ 24	2 $\times$ 24	−0.104
	fric_x0.4	24	24	2/2	2/2	2 $\times$ 24	2 $\times$ 24	−0.062
	dens_x0.5	24	24	2/2	2/2	2 $\times$ 24	2 $\times$ 24	−0.042
spoon	nominal	24	24	2/2	2/2	2 $\times$ 24	2 $\times$ 24	+0.396
	iso_x0.25	24	24	2/2	2/2	2 $\times$ 24	2 $\times$ 24	+0.250
	iso_x4.0	24	24	2/2	2/2	2 $\times$ 24	2 $\times$ 24	+0.354
	force_x0.5	24	24	2/2	2/2	2 $\times$ 24	2 $\times$ 24	+0.375
	fric_x0.4	24	24	2/2	2/2	2 $\times$ 24	2 $\times$ 24	+0.208
	dens_x0.5	24	24	2/2	2/2	2 $\times$ 24	2 $\times$ 24	+0.250
eggplant	nominal	64	64	5/5	5/5	6 $\times$ 32, 5 $\times$ 32	6 $\times$ 32, 5 $\times$ 32	+0.055
	iso_x0.25	64	64	5/5	4/4	5 $\times$ 48, 4 $\times$ 16	5 $\times$ 48, 4 $\times$ 16	+0.092
	iso_x4.0	64	64	5/5	4/4	5 $\times$ 48, 4 $\times$ 16	5 $\times$ 48, 4 $\times$ 16	+0.174
	force_x0.5	64	64	5/5	5/5	5 $\times$ 64	5 $\times$ 64	+0.106
	fric_x0.4	64	64	5/5	5/5	6 $\times$ 32, 5 $\times$ 32	6 $\times$ 32, 5 $\times$ 32	+0.077
	dens_x0.5	64	64	5/5	4/4	5 $\times$ 48, 4 $\times$ 16	5 $\times$ 48, 4 $\times$ 16	+0.128

index wraps onto the grid, an unpaired comparison silently contrasts different effective scopes — computed that way the build effect of Table 3 changes by roughly a factor of two.

And match the *pseudo-random-number lifecycle*, not only the loop structure. A per-episode re-seed and a single advancing stream can share an outer loop and still measure different things: the first makes a run one noise realisation replicated across the grid. On the protocol of Sect. 6 that choice doubled the across-run spread of a cell’s rate (0.052 to 0.109), moved the mean difference from the published rates from −0.005 to −0.038, and decided the sign of one comparison — the retraction in Sect. 1.2. Our inference server therefore reports which lifecycle it applied on every reset, and the sweep asserts it, because the failure mode is silent: a server without the fix coerces an absent seed to zero and returns plausible numbers under the wrong protocol.

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